CRANE + Factor-Adjusted Graphical Model: Full Colab Notebook

이 노트북은 한 번에 다음을 수행한다.

1. 관세청 `관세청_품목별 국가별 수출입실적(GW)` API에서 HS 01~99 전체 chapter 데이터를 수집한다.
2. API raw data를 `data_raw/customs_raw_production.csv`로 저장한다.
3. HS-국가-월 수입단가 상대가격 feature와 다음 달 이상충격 target을 만든다.
4. CRANE feature data를 `output_production/crane_features_production.csv`로 저장한다.
5. HS-월 이상점수 행렬을 만들고 Factor model + Graphical LASSO를 적합한다.
6. 그래프 기반 feature를 기존 Tweedie/Hurdle 조기경보 모델에 결합해 baseline과 비교한다.

핵심 모델:

[Z_{h,t}=\lambda_h^\top F_t+u_{h,t}, \qquad u_t \sim N(0,\Sigma_u), \qquad \Omega_u=\Sigma_u^{-1}.]

((\Omega_u)_{ij}\neq 0)이면 공통요인과 다른 HS를 통제한 뒤에도 HS (i,j) 사이에 조건부 의존성이 남는다고 해석한다.

Colab에서 오래 걸리는 작업:

- API 전체 수집: 처음 실행 시 오래 걸림. 이후 cache 재사용.
- LightGBM downstream: 데이터가 크면 오래 걸림.
- Bootstrap edge stability: 기본값 `False`; 최종 실행 때만 `True` 권장.

```
# Cell 0. Install and import packages
import importlib.util
import subprocess
import sys
import os
import time
import xml.etree.ElementTree as ET
from getpass import getpass
from pathlib import Path
from urllib.parse import unquote
import warnings

warnings.filterwarnings("ignore")


def ensure_package(import_name, pip_name=None):
    if importlib.util.find_spec(import_name) is None:
        subprocess.check_call([sys.executable, "-m", "pip", "install", "-q", pip_name or import_name])

for import_name, pip_name in [
    ("numpy", None),
    ("pandas", None),
    ("requests", None),
    ("sklearn", "scikit-learn"),
    ("statsmodels", None),
    ("networkx", None),
    ("matplotlib", None),
    ("seaborn", None),
    ("lightgbm", None),
    ("openpyxl", None),
]:
    ensure_package(import_name, pip_name)

import numpy as np
import pandas as pd
import requests
import matplotlib.pyplot as plt
import seaborn as sns
import networkx as nx
import lightgbm as lgb

from sklearn.decomposition import PCA
from sklearn.covariance import GraphicalLassoCV, GraphicalLasso
from sklearn.metrics import average_precision_score, roc_auc_score, mean_squared_error, mean_absolute_error, r2_score
from statsmodels.tsa.api import VAR

pd.set_option("display.max_columns", 250)
pd.set_option("display.width", 220)
sns.set_theme(style="whitegrid", context="notebook")

RANDOM_STATE = 123
EPS = 1e-8
```

```
for d in ["data_raw", "data_processed", "output_production", "output_factor_graphical", "figures_production", "figures_factor_graphical
    Path(d).mkdir(exist_ok=True)

print("Packages and folders ready.")
print("Working directory:", os.getcwd())
```

```
Packages and folders ready.
Working directory: /content
```

```
# Cell 1. Configuration

# 기본은 API를 다시 따는 모드다. 이미 feature 파일이 있으면 "existing_features"로 바꿔도 된다.
SOURCE_MODE = "customs_api"  # customs_api, existing_features, local_trade

# 이미 저장된 파일 재사용용 경로. customs_api 모드에서는 이 파일을 먼저 찾지 않는다.
LOCAL_FEATURE_PATHS = [
    "output_production/crane_features_production.csv",
    "crane_features_production.csv",
    "crane_features.csv",
]

LOCAL_TRADE_PATHS = [
    "data_raw/customs_raw_production.csv",
    "data_raw/customs_raw.csv",
    "customs_raw_production.csv",
    "customs_raw.csv",
    "trade.csv",
    "trade.xlsx",
]

# Customs API settings
START_YYYYMM = "202001"
END_YYYYMM = "202412"
HS_SCOPE = "all_chapters"  # all_chapters, selected
HS_CODES = [f"{i:02d}" for i in range(1, 100)]
COUNTRY_CODES = [None]  # None: all countries if API accepts omitted cntyCd
CUSTOMS_CACHE_DIR = "data_raw/customs_api_cache"
OVERWRITE_CUSTOMS_CACHE = False
CUSTOMS_SLEEP_SEC = 0.25
CUSTOMS_MAX_RETRIES = 3
CUSTOMS_MAX_PAGES_PER_JOB = None

# CRANE feature settings
ANOMALY_K = 1.5
MIN_COUNTRIES_PER_HS_MONTH = 3
TRAIN_END = "2023-01-01"
VALID_END = "2024-01-01"
MATERIAL_QUANTILE = 0.90
K_VALUES = [10, 30, 50, 100]

# Graph model settings
HS_AGG_METHOD = "value_weighted_abs"
MAX_HS_NODES = 120
MIN_MONTH_COVERAGE = 24
FILL_MISSING_METHOD = "zero_after_standardize"
N_FACTORS = 5
MAX_FACTORS_TO_PLOT = 8
VAR_MAX_LAG = 3
VAR_SELECTED_LAG = None
EDGE_ABS_PARTIAL_CORR_THRESHOLD = 0.05
TOP_EDGES_TO_PLOT = 180
RUN_BOOTSTRAP_STABILITY = False  # Colab-safe. 최종 발표용만 True 권장.
BOOTSTRAP_B = 50
BOOTSTRAP_BLOCK_SIZE = 6
EDGE_STABILITY_THRESHOLD = 0.50

# Downstream prediction settings
RUN_DOWNSTREAM_LIGHTGBM = True

# Plot settings
SAVE_FIGURES = True
FIG_DPI = 140

print("Configuration loaded.")
```

```
Configuration loaded.
```

```
# Cell 2. General helper functions

def read_table_auto(path_or_url):
    path_or_url = str(path_or_url)
```

```
    lower = path_or_url.lower()
    if lower.startswith("http://") or lower.startswith("https://"):
        if lower.endswith((".xlsx", ".xls")):
            return pd.read_excel(path_or_url)
        return pd.read_csv(path_or_url)
    path = Path(path_or_url)
    if not path.exists():
        raise FileNotFoundError(path)
    if lower.endswith((".xlsx", ".xls")):
        return pd.read_excel(path)
    if lower.endswith(".parquet"):
        return pd.read_parquet(path)
    if lower.endswith(".zip"):
        return pd.read_csv(path, compression="zip")
    try:
        return pd.read_csv(path, encoding="utf-8-sig")
    except UnicodeDecodeError:
        return pd.read_csv(path, encoding="cp949")


def first_existing(paths):
    for p in paths:
        if Path(p).exists():
            return p
    return None


def clean_numeric(series):
    return pd.to_numeric(
        series.astype(str)
        .str.replace(",", "", regex=False)
        .str.replace(" ", "", regex=False)
        .str.strip(),
        errors="coerce",
    )


def pick_column(df, candidates):
    lower_map = {str(c).lower().strip(): c for c in df.columns}
    for cand in candidates:
        key = cand.lower().strip()
        if key in lower_map:
            return lower_map[key]
    return None


def robust_mad(x):
    arr = pd.Series(x).dropna().to_numpy(dtype=float)
    if len(arr) == 0:
        return np.nan
    med = np.nanmedian(arr)
    return np.nanmedian(np.abs(arr - med))


def save_fig(path, min_bytes=3000):
    path = Path(path)
    if SAVE_FIGURES:
        plt.tight_layout()
        plt.savefig(path, dpi=FIG_DPI, bbox_inches="tight")
    plt.show()
    if SAVE_FIGURES:
        if not path.exists() or path.stat().st_size < min_bytes:
            print(f"WARNING: figure may be too small or not saved correctly: {path}")
        else:
            print("saved figure:", path, "bytes=", path.stat().st_size)


def pct_rank_score(x):
    return pd.Series(x).rank(pct=True).fillna(0.5).to_numpy()


def safe_zscore_matrix(mat):
    out = mat.copy().astype(float)
    mu = out.mean(axis=0, skipna=True)
    sd = out.std(axis=0, skipna=True).replace(0, np.nan)
    out = (out - mu) / sd
    return out, mu, sd

print("Helpers ready.")
```

```
Helpers ready.
```

```
# Cell 3. Customs API functions

def make_yearly_periods(start_yyyymm, end_yyyymm):
    start_year = int(start_yyyymm[:4])
    end_year = int(end_yyyymm[:4])
    periods = []
    for year in range(start_year, end_year + 1):
        start_month = int(start_yyyymm[4:6]) if year == start_year else 1
        end_month = int(end_yyyymm[4:6]) if year == end_year else 12
        periods.append((f"{year}{start_month:02d}", f"{year}{end_month:02d}"))
    return periods


def resolve_hs_codes():
    if HS_SCOPE == "all_chapters":
        return [f"{i:02d}" for i in range(1, 100)]
    if HS_SCOPE == "selected":
        return [str(x).zfill(2) if str(x).isdigit() and len(str(x)) == 1 else str(x) for x in HS_CODES]
    raise ValueError("HS_SCOPE must be 'all_chapters' or 'selected'.")


def safe_cache_name(hs_code, country_code, start_yyyymm, end_yyyymm):
    country = country_code if country_code not in (None, "") else "ALL"
    return f"customs_hs_{hs_code}_country_{country}_{start_yyyymm}_{end_yyyymm}.csv"


def call_customs_api_xml(service_key, start_yyyymm, end_yyyymm, hs_code=None, country_code=None, page_no=1, num_rows=10000):
    base_url = "https://apis.data.go.kr/1220000/nitemtrade/getNitemtradeList"
    params = {
        "serviceKey": service_key,
        "strtYymm": start_yyyymm,
        "endYymm": end_yyyymm,
        "pageNo": page_no,
        "numOfRows": num_rows,
    }
    if hs_code:
        params["hsSgn"] = str(hs_code)
    if country_code:
        params["cntyCd"] = str(country_code)
    response = requests.get(base_url, params=params, timeout=90)
    if response.status_code != 200:
        raise RuntimeError(f"Customs HTTP {response.status_code}: {response.text[:500]}")
    return response.text


def parse_customs_xml(xml_text):
    root = ET.fromstring(xml_text)
    result_code = root.findtext(".//resultCode")
    result_msg = root.findtext(".//resultMsg")
    if result_code not in (None, "00"):
        raise RuntimeError(f"Customs API resultCode={result_code}, resultMsg={result_msg}")
    total_text = root.findtext(".//totalCount")
    total_count = int(total_text) if total_text else None
    rows = [{child.tag: child.text for child in item} for item in root.findall(".//item")]
    return pd.DataFrame(rows), total_count


def collect_customs_api(
    service_key,
    hs_codes,
    start_yyyymm,
    end_yyyymm,
    country_codes=(None,),
    cache_dir=CUSTOMS_CACHE_DIR,
    overwrite_cache=OVERWRITE_CUSTOMS_CACHE,
    sleep_sec=CUSTOMS_SLEEP_SEC,
    max_retries=CUSTOMS_MAX_RETRIES,
    max_pages_per_job=CUSTOMS_MAX_PAGES_PER_JOB,
):
    periods = make_yearly_periods(start_yyyymm, end_yyyymm)
    cache_dir = Path(cache_dir)
    cache_dir.mkdir(parents=True, exist_ok=True)
    frames = []
    failures = []

    for hs in hs_codes:
        for country in country_codes:
            for start, end in periods:
                cache_path = cache_dir / safe_cache_name(hs, country, start, end)
                if cache_path.exists() and not overwrite_cache:
                    try:
                        cached = read_table_auto(cache_path)
```

```
                    frames.append(cached)
                    print(f"Cache hit: hs={hs}, country={country or 'ALL'}, {start}-{end}, rows={len(cached)}")
                    continue
                except Exception as err:
                    print("Cache read failed, refetching:", cache_path, err)

            page = 1
            job_pages = []
            while True:
                try:
                    df_page = pd.DataFrame()
                    total_count = None
                    last_error = None
                    for attempt in range(1, max_retries + 1):
                        try:
                            xml_text = call_customs_api_xml(service_key, start, end, hs, country, page_no=page)
                            df_page, total_count = parse_customs_xml(xml_text)
                            break
                        except Exception as err:
                            last_error = err
                            wait = sleep_sec * attempt
                            print(f"Retry {attempt}/{max_retries}: hs={hs}, {start}-{end}, page={page}, error={err}")
                            time.sleep(wait)
                    else:
                        raise RuntimeError(last_error)

                    if df_page.empty:
                        break

                    df_page["query_hs_code"] = hs
                    df_page["query_country_code"] = country if country else ""
                    df_page["query_start_yyyymm"] = start
                    df_page["query_end_yyyymm"] = end
                    df_page["query_page_no"] = page
                    job_pages.append(df_page)
                    print(f"Customs hs={hs}, country={country or 'ALL'}, {start}-{end}, page={page}, rows={len(df_page)}, total={tc

                    if max_pages_per_job is not None and page >= max_pages_per_job:
                        print("Stopped by CUSTOMS_MAX_PAGES_PER_JOB for debugging.")
                        break
                    if total_count is None or page * 10000 >= total_count:
                        break
                    page += 1
                    time.sleep(sleep_sec)
                except Exception as err:
                    failures.append({
                        "hs_code": hs,
                        "country_code": country,
                        "start_yyyymm": start,
                        "end_yyyymm": end,
                        "page": page,
                        "error": str(err),
                    })
                    print("Customs failure:", failures[-1])
                    break

            if job_pages:
                job_df = pd.concat(job_pages, ignore_index=True)
                job_df.to_csv(cache_path, index=False, encoding="utf-8-sig")
                frames.append(job_df)

    raw = pd.concat(frames, ignore_index=True) if frames else pd.DataFrame()
    failed = pd.DataFrame(failures)
    raw.to_csv("data_raw/customs_raw_production.csv", index=False, encoding="utf-8-sig")
    failed.to_csv("data_raw/customs_failed_jobs_production.csv", index=False, encoding="utf-8-sig")
    return raw, failed

print("Customs API functions ready.")
```

```
Customs API functions ready.
```

```
# Cell 4. Standardize Customs/API/local trade data

def standardize_trade_dataframe(df):
    """Standardize Customs/API/local trade data to HS-country-month cells."""
    out = df.copy()
    out.columns = [str(c).strip() for c in out.columns]

    col_date = pick_column(out, ["date", "ym", "year_month", "yearmonth", "time", "TIME", "period"])
    col_year = pick_column(out, ["year", "YEAR"])
    col_hs = pick_column(out, ["hs_code", "hsCd", "hscd", "hs", "hsSgn", "cmdCode", "commodity_code"])
    col_country = pick_column(out, ["country_code", "statCd", "cntyCd", "partnerCode", "partner_code", "country"])
```

```
    col_country_name = pick_column(out, ["country_name", "statCdCntnKor1", "partnerDesc", "partner_name"])
    col_hs_name = pick_column(out, ["hs_name", "statKor", "cmdDesc", "commodity"])
    col_value = pick_column(out, ["import_value", "impDlr", "importValue", "tradeValue", "cifvalue", "value"])
    col_weight = pick_column(out, ["import_weight", "impWgt", "netWgt", "net_weight", "weight"])

    required = {"HS": col_hs, "country": col_country, "value": col_value, "weight": col_weight}
    missing = [k for k, v in required.items() if v is None]
    if missing:
        raise KeyError(f"Required columns could not be identified: {missing}. Available columns: {list(out.columns)[:80]}")

    if col_date is not None:
        date_raw = out[col_date].astype(str).str.strip()
    elif col_year is not None:
        date_raw = out[col_year].astype(str).str.strip()
    else:
        raise KeyError("Could not identify date/year_month column.")

    date_raw = date_raw.str.replace(".", "", regex=False).str.replace("-", "", regex=False).str.replace("/", "", regex=False)
    ym = date_raw.str.extract(r"(₩d{6})", expand=False)
    out["date"] = pd.to_datetime(ym + "01", format="%Y%m%d", errors="coerce")

    out["hs_code"] = out[col_hs].astype(str).str.strip().str.replace(r"₩.0$", "", regex=True)
    out["country_code"] = out[col_country].astype(str).str.strip().str.replace(r"₩.0$", "", regex=True)
    out["country_name"] = out[col_country_name].astype(str).str.strip() if col_country_name else out["country_code"]
    out["hs_name"] = out[col_hs_name].astype(str).str.strip() if col_hs_name else out["hs_code"]
    out["import_value"] = clean_numeric(out[col_value])
    out["import_weight"] = clean_numeric(out[col_weight])

    out = out[
        out["date"].notna()
        & out["hs_code"].notna()
        & out["country_code"].notna()
        & (out["hs_code"] != "-")
        & (out["country_code"] != "-")
        & out["import_value"].notna()
        & out["import_weight"].notna()
        & (out["import_value"] > 0)
        & (out["import_weight"] > 0)
    ].copy()

    agg = (
        out.groupby(["date", "hs_code", "country_code"], as_index=False)
        .agg(
            country_name=("country_name", "first"),
            hs_name=("hs_name", "first"),
            import_value=("import_value", "sum"),
            import_weight=("import_weight", "sum"),
        )
        .sort_values(["hs_code", "country_code", "date"])
    )
    agg["year_month"] = agg["date"].dt.strftime("%Y%m")
    agg["year"] = agg["date"].dt.year
    agg["month"] = agg["date"].dt.month
    agg["unit_price"] = agg["import_value"] / agg["import_weight"]
    agg["log_price"] = np.log(agg["unit_price"])
    agg = agg.replace([np.inf, -np.inf], np.nan).dropna(subset=["log_price"])
    return agg.reset_index(drop=True)


def diagnose_hs_scope(df, label):
    if "hs_code" not in df.columns:
        return
    hs = df["hs_code"].astype(str).str.replace(r"₩.0$", "", regex=True).str.zfill(4)
    chapters = hs.str[:2]
    summary = pd.DataFrame({
        "metric": ["rows", "unique_hs_codes", "unique_hs_chapters", "min_chapter", "max_chapter"],
        "value": [len(df), hs.nunique(), chapters.nunique(), chapters.min(), chapters.max()],
    })
    print(f"HS scope diagnostic: {label}")
    display(summary)
    display(chapters.value_counts().sort_index().rename_axis("hs_chapter").reset_index(name="rows"))

print("Standardization functions ready.")
```

```
Standardization functions ready.
```

```
# Cell 5. Load data from API or existing file
# customs_api 모드에서는 local CSV를 먼저 찾지 않고 API를 실행한다.

if SOURCE_MODE == "customs_api":
    active_hs_codes = resolve_hs_codes()
    print(
```

```
        f"Customs API production collection: HS_SCOPE={HS_SCOPE}, "
        f"HS jobs={len(active_hs_codes)}, period={START_YYYYMM}-{END_YYYYMM}. "
        "Cached jobs are reused unless OVERWRITE_CUSTOMS_CACHE=True."
    )
    raw_key = getpass("Public Data Portal service key: ").strip()
    raw = pd.DataFrame()
    last_error = None
    for key in [unquote(raw_key), raw_key]:
        try:
            raw, failed = collect_customs_api(
                key,
                hs_codes=active_hs_codes,
                start_yyyymm=START_YYYYMM,
                end_yyyymm=END_YYYYMM,
                country_codes=COUNTRY_CODES,
            )
            if not raw.empty:
                break
        except Exception as err:
            last_error = err
            print("Key attempt failed:", err)
    if raw.empty:
        raise RuntimeError(f"Customs API collection failed: {last_error}")
    crane_or_trade = standardize_trade_dataframe(raw)
    print("Collected and standardized trade data from Customs API.")

elif SOURCE_MODE == "existing_features":
    feature_path = first_existing(LOCAL_FEATURE_PATHS)
    if feature_path is None:
        raise FileNotFoundError(f"No existing feature file found in: {LOCAL_FEATURE_PATHS}")
    crane_or_trade = read_table_auto(feature_path)
    crane_or_trade["date"] = pd.to_datetime(crane_or_trade["date"], errors="coerce")
    if "next_date" in crane_or_trade.columns:
        crane_or_trade["next_date"] = pd.to_datetime(crane_or_trade["next_date"], errors="coerce")
    print(f"Loaded existing feature file: {feature_path}")

elif SOURCE_MODE == "local_trade":
    trade_path = first_existing(LOCAL_TRADE_PATHS)
    if trade_path is None:
        raise FileNotFoundError(f"No local trade file found in: {LOCAL_TRADE_PATHS}")
    raw_local = read_table_auto(trade_path)
    crane_or_trade = standardize_trade_dataframe(raw_local)
    print(f"Loaded raw trade file: {trade_path}")

else:
    raise ValueError("SOURCE_MODE must be customs_api, existing_features, or local_trade.")

print("loaded shape:", crane_or_trade.shape)
display(crane_or_trade.head())
diagnose_hs_scope(crane_or_trade, "loaded input")

# Save standardized trade frame as a stable intermediate.
if not {"rel_z", "future_net_anomaly_amount"}.issubset(crane_or_trade.columns):
    crane_or_trade.to_csv("data_processed/trade_clean_production.csv", index=False, encoding="utf-8-sig")
    print("Saved standardized trade frame: data_processed/trade_clean_production.csv")
```

```
Customs API production collection: HS_SCOPE=all_chapters, HS jobs=99, period=202001-202412. Cached jobs are reused unless OVERWRITE_CU
Public Data Portal service key: · · · · · · · · · · ·
Customs hs=01, country=ALL, 202001-202012, page=1, rows=599, total=None
Customs hs=01, country=ALL, 202101-202112, page=1, rows=696, total=None
Customs hs=01, country=ALL, 202201-202212, page=1, rows=653, total=None
Customs hs=01, country=ALL, 202301-202312, page=1, rows=671, total=None
Customs hs=01, country=ALL, 202401-202412, page=1, rows=642, total=None
Customs hs=02, country=ALL, 202001-202012, page=1, rows=901, total=None
Customs hs=02, country=ALL, 202101-202112, page=1, rows=905, total=None
Customs hs=02, country=ALL, 202201-202212, page=1, rows=914, total=None
Customs hs=02, country=ALL, 202301-202312, page=1, rows=875, total=None
Customs hs=02, country=ALL, 202401-202412, page=1, rows=983, total=None
Customs hs=03, country=ALL, 202001-202012, page=1, rows=3520, total=None
Customs hs=03, country=ALL, 202101-202112, page=1, rows=3541, total=None
Customs hs=03, country=ALL, 202201-202212, page=1, rows=3566, total=None
Customs hs=03, country=ALL, 202301-202312, page=1, rows=3687, total=None
Customs hs=03, country=ALL, 202401-202412, page=1, rows=3653, total=None
Customs hs=04, country=ALL, 202001-202012, page=1, rows=2296, total=None
Customs hs=04, country=ALL, 202101-202112, page=1, rows=2446, total=None
Customs hs=04, country=ALL, 202201-202212, page=1, rows=2390, total=None
Customs hs=04, country=ALL, 202301-202312, page=1, rows=2416, total=None
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Collected and standardized trade data from Customs API.
loaded shape: (1409337, 12)
```

	date	hs_code	country_code	country_name	hs_name	import_value	import_weight	year_month	year	month	unit_price
0	2020-06-01	0101	AR	아르헨티나	살아 있는 말·당나귀·노새·버새	244500	4000	202006	2020	6	61.125000
1	2021-03-01	0101	AR	아르헨티나	살아 있는 말·당나귀·노새·버새	434300	4000	202103	2021	3	108.575000
2	2021-04-01	0101	AR	아르헨티나	살아 있는 말·당나귀·노새·버새	266700	4000	202104	2021	4	66.675000
3	2021-08-01	0101	AR	아르헨티나	살아 있는 말·당나귀·노새·버새	88600	1200	202108	2021	8	73.833333
4	2022-05-01	0101	AR	아르헨티나	살아 있는 말·당나귀·노새·버새	673000	6750	202205	2022	5	99.703704

```
# Cell 6. CRANE feature engineering

def leave_one_out_median_series(s):
    vals = s.to_numpy(dtype=float)
    n = len(vals)
    if n <= 2:
        return pd.Series(np.nan, index=s.index)
    result = np.empty(n, dtype=float)
    for i in range(n):
        result[i] = np.nanmedian(np.delete(vals, i))
    return pd.Series(result, index=s.index)


def isolation_score_series(s, eps=EPS):
    vals = s.to_numpy(dtype=float)
    n = len(vals)
    if n <= 2:
        return pd.Series(np.nan, index=s.index)
    scale = robust_mad(vals)
    if not np.isfinite(scale) or scale <= 0:
        scale = np.nanstd(vals)
    scale = max(float(scale), eps)
    dist = np.abs(vals[:, None] - vals[None, :])
    sim = np.exp(-dist / scale)
    np.fill_diagonal(sim, np.nan)
    return pd.Series(1.0 - np.nanmean(sim, axis=1), index=s.index)


def add_crane_features(trade, k=ANOMALY_K, min_countries=MIN_COUNTRIES_PER_HS_MONTH):
    out = trade.copy()
    if "unit_price" not in out.columns or "log_price" not in out.columns:
        out = standardize_trade_dataframe(out)
    out["date"] = pd.to_datetime(out["date"])
    out["hs_code"] = out["hs_code"].astype(str)
    out["country_code"] = out["country_code"].astype(str)
    out = out.sort_values(["hs_code", "country_code", "date"]).reset_index(drop=True)

    group_cols = ["hs_code", "date"]
    grp = out.groupby(group_cols)["log_price"]
    out["ht_country_count"] = grp.transform("count")
    out["ht_median_log_price"] = grp.transform("median")
    out["ht_mad_log_price"] = grp.transform(lambda x: robust_mad(x)).clip(lower=EPS)
    out["rel_z"] = (out["log_price"] - out["ht_median_log_price"]) / out["ht_mad_log_price"]

    rank = grp.rank(method="average")
    out["price_rank_pct"] = np.where(out["ht_country_count"] > 1, (rank - 1) / (out["ht_country_count"] - 1), np.nan)
    out.loc[out["ht_country_count"] < min_countries, ["rel_z", "price_rank_pct"]] = np.nan

    out["loo_median_log_price"] = (
        out.groupby(group_cols)["log_price"].apply(leave_one_out_median_series).reset_index(level=group_cols, drop=True)
    )
    out["counterfactual_dev"] = out["log_price"] - out["loo_median_log_price"]
    out["counterfactual_dev_scaled"] = out["counterfactual_dev"] / out["ht_mad_log_price"]
    out["network_isolation_score"] = (
        out.groupby(group_cols)["log_price"].apply(isolation_score_series).reset_index(level=group_cols, drop=True)
    )

    out["total_hs_month_import"] = out.groupby(group_cols)["import_value"].transform("sum")
    out["import_share_country"] = out["import_value"] / out["total_hs_month_import"]
    out["hhi_hs_month"] = out.groupby(group_cols)["import_share_country"].transform(lambda x: float(np.sum(np.square(x))))
    out["concentration_risk"] = out["import_share_country"] * out["hhi_hs_month"]

    panel = out.groupby(["hs_code", "country_code"], group_keys=False)
    out["lag_date"] = panel["date"].shift(1)
    out["price_return"] = panel["log_price"].diff()
    out["rank_shift"] = out["price_rank_pct"] - panel["price_rank_pct"].shift(1)
    continuous_lag = out["date"].eq(out["lag_date"] + pd.DateOffset(months=1))
    out.loc[~continuous_lag, ["price_return", "rank_shift"]] = np.nan

    out["abs_rank_shift"] = out["rank_shift"].abs()
    out["import_growth"] = panel["import_value"].pct_change()
    out["weight_growth"] = panel["import_weight"].pct_change()
    out["log_import_value"] = np.log(out["import_value"])
    out["log_import_weight"] = np.log(out["import_weight"])
    out["abs_rel_z"] = out["rel_z"].abs()

    out["anomaly_abs"] = (out["rel_z"].abs() > k).astype(float)
    out["anomaly_up"] = (out["rel_z"] > k).astype(float)
    out["anomaly_down"] = (out["rel_z"] < -k).astype(float)

    for col in ["anomaly_abs", "anomaly_up", "anomaly_down", "abs_rel_z"]:
        out[f"{col}_roll3"] = panel[col].transform(lambda x: x.shift(1).rolling(3, min_periods=1).mean())
```

```
    out["price_return_roll3"] = panel["price_return"].transform(lambda x: x.shift(1).rolling(3, min_periods=2).mean())
    out["price_volatility_roll3"] = panel["price_return"].transform(lambda x: x.shift(1).rolling(3, min_periods=2).std())
    out["import_growth_roll3"] = panel["import_growth"].transform(lambda x: x.shift(1).rolling(3, min_periods=2).mean())

    out["return_sign"] = np.sign(out["price_return"])

    def majority_sign(x):
        nonzero = x.dropna()
        nonzero = nonzero[nonzero != 0]
        if len(nonzero) == 0:
            return np.nan
        return 1.0 if (nonzero > 0).mean() >= 0.5 else -1.0

    out["majority_sign"] = out.groupby(group_cols)["return_sign"].transform(majority_sign)
    out["sync_score"] = out.groupby(group_cols)["return_sign"].transform(
        lambda x: np.nan if x.dropna().empty else np.nanmean(x == out.loc[x.index, "majority_sign"])
    )
    out["idiosyncratic_current_score"] = out["abs_rel_z"] * (1 - out["sync_score"].fillna(0))

    stats = (
        out.groupby(group_cols, as_index=False)
        .agg(
            ht_median_for_shift=("log_price", "median"),
            ht_mad_for_shift=("log_price", robust_mad),
            ht_total_import_value=("import_value", "sum"),
        )
        .sort_values(["hs_code", "date"])
    )
    stats["ht_median_return"] = stats.groupby("hs_code")["ht_median_for_shift"].diff()
    stats["ht_mad_return"] = stats.groupby("hs_code")["ht_mad_for_shift"].diff()
    stats["total_import_return"] = stats.groupby("hs_code")["ht_total_import_value"].pct_change()
    stats["price_distribution_shift"] = (
        stats["ht_median_return"].abs() / (stats["ht_mad_for_shift"].abs() + EPS)
        + stats["ht_mad_return"].abs() / (stats["ht_mad_for_shift"].abs() + EPS)
    )
    out = out.merge(
        stats[["hs_code", "date", "ht_median_return", "ht_mad_return", "total_import_return", "price_distribution_shift"]],
        on=group_cols,
        how="left",
    )

    out["month_sin"] = np.sin(2 * np.pi * out["month"] / 12)
    out["month_cos"] = np.cos(2 * np.pi * out["month"] / 12)
    return out.replace([np.inf, -np.inf], np.nan)


def add_future_targets(df, k=ANOMALY_K):
    out = df.copy().sort_values(["hs_code", "country_code", "date"]).reset_index(drop=True)
    panel = out.groupby(["hs_code", "country_code"], group_keys=False)
    out["next_date"] = panel["date"].shift(-1)
    out["next_rel_z"] = panel["rel_z"].shift(-1)
    out["next_price_rank_pct"] = panel["price_rank_pct"].shift(-1)
    if "network_isolation_score" in out.columns:
        out["next_network_isolation_score"] = panel["network_isolation_score"].shift(-1)
    out["has_next_month"] = out["next_date"].eq(out["date"] + pd.DateOffset(months=1))

    excess_abs = np.maximum(out["next_rel_z"].abs() - k, 0)
    excess_up = np.maximum(out["next_rel_z"] - k, 0)
    excess_down = np.maximum(-out["next_rel_z"] - k, 0)

    out["future_abs_anomaly"] = np.where(out["has_next_month"], excess_abs, np.nan)
    out["future_up_anomaly"] = np.where(out["has_next_month"], excess_up, np.nan)
    out["future_down_anomaly"] = np.where(out["has_next_month"], excess_down, np.nan)
    out["future_net_anomaly_amount"] = out["import_value"] * out["future_abs_anomaly"]
    out["future_up_anomaly_amount"] = out["import_value"] * out["future_up_anomaly"]
    out["future_down_anomaly_amount"] = out["import_value"] * out["future_down_anomaly"]
    return out

needed = {"rel_z", "future_net_anomaly_amount", "import_value", "import_weight", "hs_code", "country_code", "date"}
if needed.issubset(set(crane_or_trade.columns)):
    crane = crane_or_trade.copy()
    print("Existing CRANE features detected and reused.")
else:
    trade_clean = standardize_trade_dataframe(crane_or_trade) if "unit_price" not in crane_or_trade.columns else crane_or_trade.copy()
    crane = add_crane_features(trade_clean)
    crane = add_future_targets(crane)
    print("CRANE features created from trade data.")

if "future_net_anomaly_amount" not in crane.columns:
    crane = add_future_targets(crane)
```

```
crane["date"] = pd.to_datetime(crane["date"])
crane["hs_code"] = crane["hs_code"].astype(str)
crane["country_code"] = crane["country_code"].astype(str)
crane.to_csv("output_production/crane_features_production.csv", index=False, encoding="utf-8-sig")
print("Saved CRANE feature file: output_production/crane_features_production.csv")
print("crane shape:", crane.shape)
display(crane.head())
diagnose_hs_scope(crane, "CRANE feature frame")
```

```
CRANE features created from trade data.
Saved CRANE feature file: output_production/crane_features_production.csv
crane shape: (1409337, 65)
```

	date	hs_code	country_code	country_name	hs_name	import_value	import_weight	year_month	year	month	unit_price
0	2020-06-01	0101	AR	아르헨티나	살아 있는 말 · 당나귀 · 노새 · 버새	244500	4000	202006	2020	6	61.125000
1	2021-03-01	0101	AR	아르헨티나	살아 있는 말 · 당나귀 · 노새 · 버새	434300	4000	202103	2021	3	108.575000
2	2021-04-01	0101	AR	아르헨티나	살아 있는 말 · 당나귀 · 노새 · 버새	266700	4000	202104	2021	4	66.675000
3	2021-08-01	0101	AR	아르헨티나	살아 있는 말 · 당나귀 · 노새 · 버새	88600	1200	202108	2021	8	73.833333
4	2022-05-01	0101	AR	아르헨티나	살아 있는 말 · 당나귀 · 노새 · 버새	673000	6750	202205	2022	5	99.703704

```
HS scope diagnostic: CRANE feature frame
```

	metric	value
0	rows	1409337
1	unique_hs_codes	1229
2	unique_hs_chapters	97
3	min_chapter	01
4	max_chapter	99

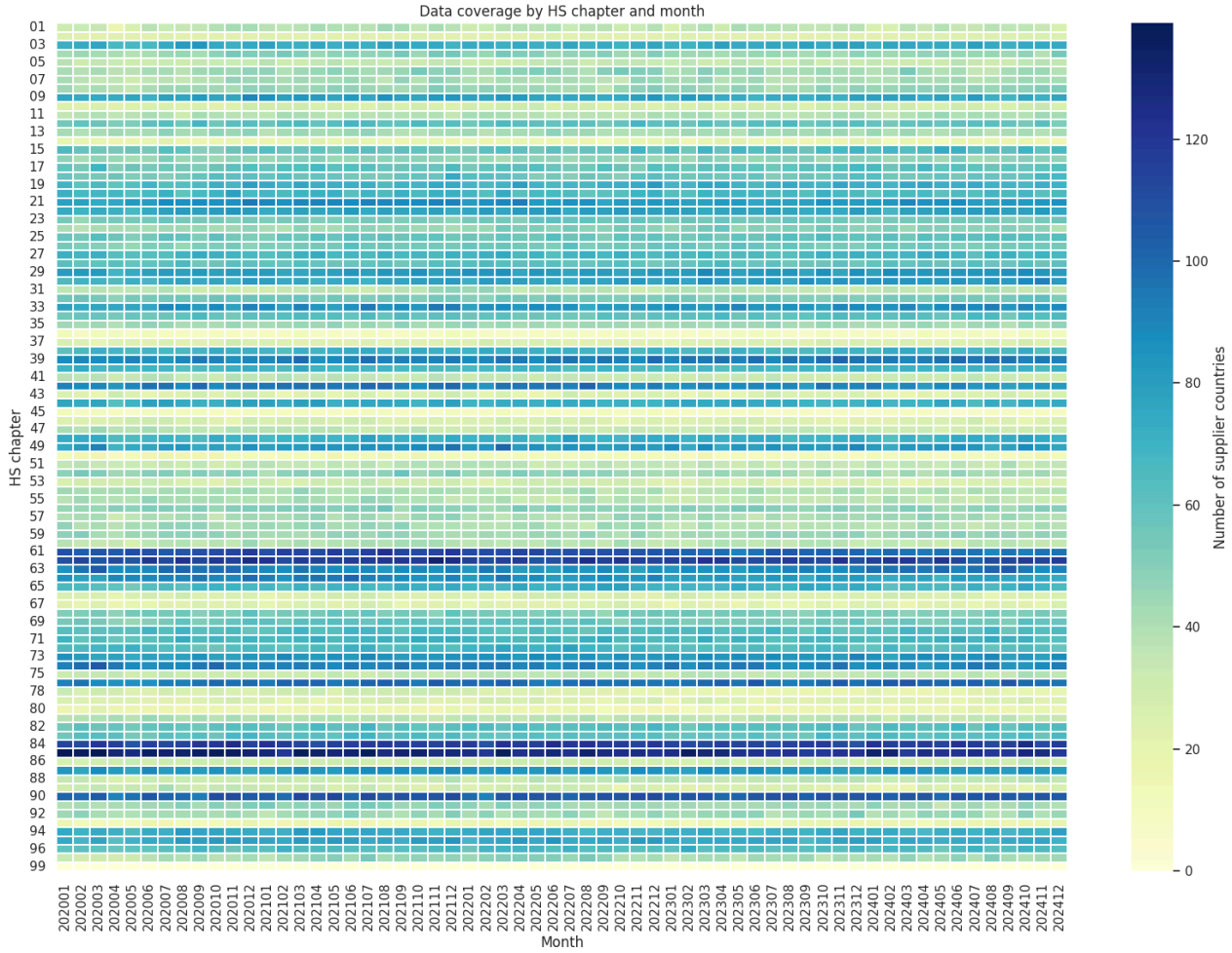
	hs_chapter	rows
0	01	2488
1	02	3984
2	03	12980
3	04	8929
4	05	3483
...	...	...
92	94	14291
93	95	12946
94	96	22062
95	97	5545
96	99	6

97 rows × 2 columns

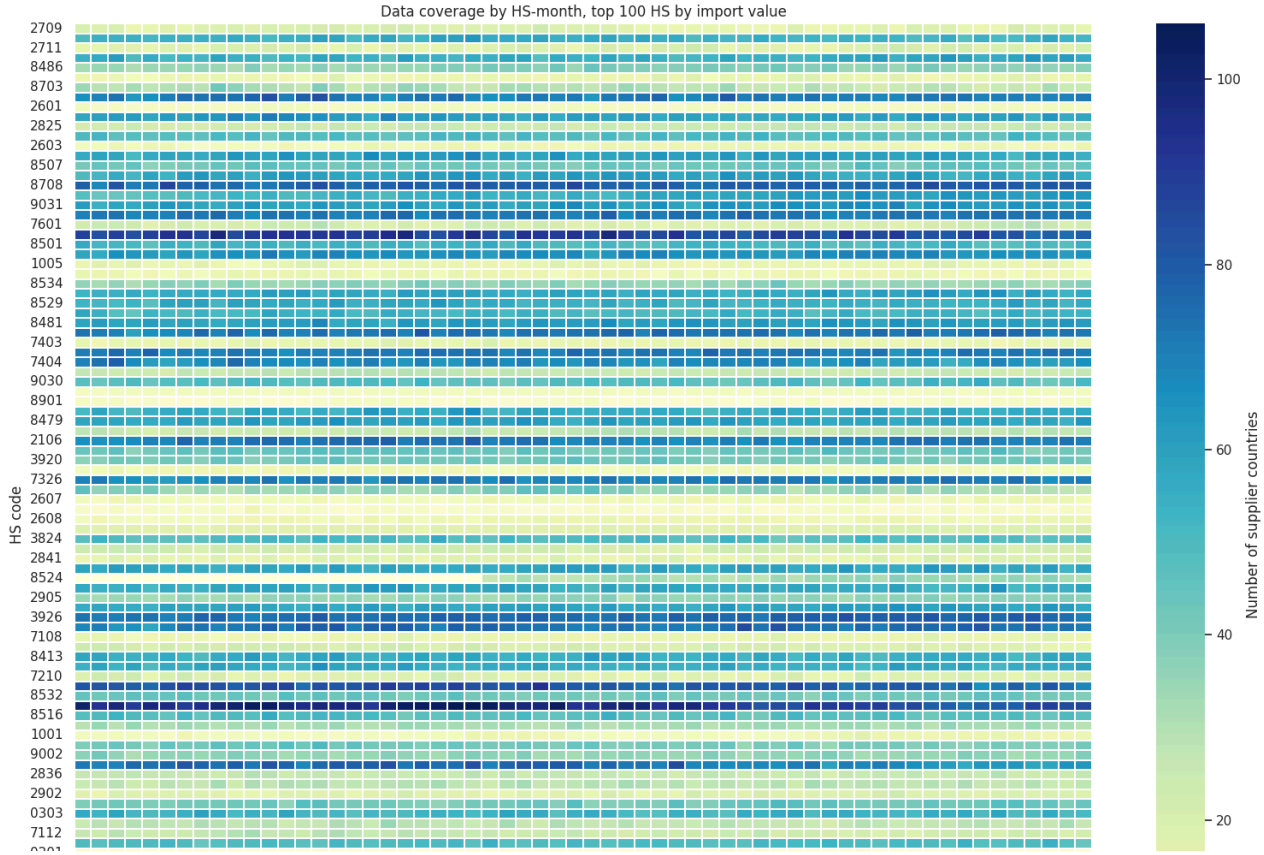
```
# Cell 7. Safe visual diagnostics

def quick_data_check(df):
    print("Data shape:", df.shape)
    print("Date range:", df["date"].min(), "to", df["date"].max())
    print("Number of months:", df["date"].nunique())
    print("Unique HS codes:", df["hs_code"].nunique())
    print("Unique HS chapters:", df["hs_code"].astype(str).str.zfill(4).str[:2].nunique())
    print("Unique countries:", df["country_code"].nunique())
    y = df["future_net_anomaly_amount"]
    print("Missing target ratio:", y.isna().mean())
    print("Zero target ratio:", (y == 0).mean())
```

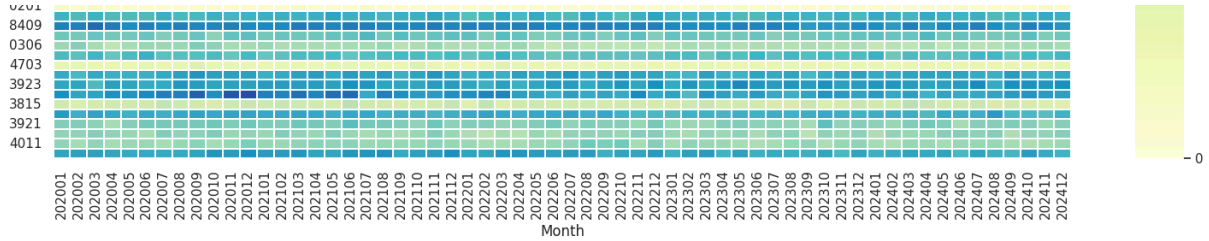
```
    print("Positive target ratio:", (y > 0).mean())


def plot_chapter_coverage_heatmap(df):
    plot_df = df.copy()
    plot_df["hs_chapter"] = plot_df["hs_code"].astype(str).str.replace(r"₩.0$", "", regex=True).str.zfill(4).str[:2]
    coverage = plot_df.pivot_table(index="hs_chapter", columns="year_month", values="country_code", aggfunc="nunique", fill_value=0)
    coverage = coverage.sort_index()
    plt.figure(figsize=(16, max(8, 0.12 * coverage.shape[0])))
    sns.heatmap(coverage, cmap="YlGnBu", linewidths=0.15, linecolor="white", cbar_kws={"label": "Number of supplier countries"})
    plt.title("Data coverage by HS chapter and month")
    plt.xlabel("Month")
    plt.ylabel("HS chapter")
    save_fig("figures_production/01a_chapter_coverage_heatmap.png")


def plot_top_hs_coverage_heatmap(df, top_n=100):
    top_hs = df.groupby("hs_code")["import_value"].sum().sort_values(ascending=False).head(top_n).index
    plot_df = df[df["hs_code"].isin(top_hs)].copy()
    coverage = plot_df.pivot_table(index="hs_code", columns="year_month", values="country_code", aggfunc="nunique", fill_value=0)
    ordered_hs = [h for h in top_hs if h in coverage.index]
    coverage = coverage.loc[ordered_hs]
    plt.figure(figsize=(16, max(8, 0.13 * coverage.shape[0])))
    sns.heatmap(coverage, cmap="YlGnBu", linewidths=0.1, linecolor="white", cbar_kws={"label": "Number of supplier countries"})
    plt.title(f"Data coverage by HS-month, top {top_n} HS by import value")
    plt.xlabel("Month")
    plt.ylabel("HS code")
    save_fig(f"figures_production/01b_top{top_n}_hs_coverage_heatmap.png")


def plot_target_distribution(df, sample_n=300000):
    y = df["future_net_anomaly_amount"].dropna()
    y_plot = y.sample(sample_n, random_state=RANDOM_STATE) if len(y) > sample_n else y
    fig, axes = plt.subplots(1, 2, figsize=(14, 4.8))
    axes[0].bar(["zero", "positive"], [(y == 0).mean(), (y > 0).mean()])
    axes[0].set_title("Target zero/positive ratio")
    axes[0].set_ylim(0, 1)
    axes[0].set_ylabel("Share")
    pos = y_plot[y_plot > 0]
    if len(pos) > 0:
        sns.histplot(np.log1p(pos), bins=50, ax=axes[1])
        axes[1].set_title(f"Positive target distribution, log1p scale₩nsample n={len(pos):,}")
    axes[1].set_xlabel("log1p(future anomaly amount)")
    save_fig("figures_production/02_target_distribution.png")

quick_data_check(crane)
plot_chapter_coverage_heatmap(crane)
plot_top_hs_coverage_heatmap(crane, top_n=100)
plot_target_distribution(crane, sample_n=300000)

display(crane["future_net_anomaly_amount"].describe(percentiles=[0.5, 0.75, 0.9, 0.95, 0.99]))
```

```
Data shape: (1409337, 65)
Date range: 2020-01-01 00:00:00 to 2024-12-01 00:00:00
Number of months: 60
Unique HS codes: 1229
Unique HS chapters: 97
Unique countries: 249
Missing target ratio: 0.21512101080153292
Zero target ratio: 0.55642546814566
Positive target ratio: 0.2284535210528071
```

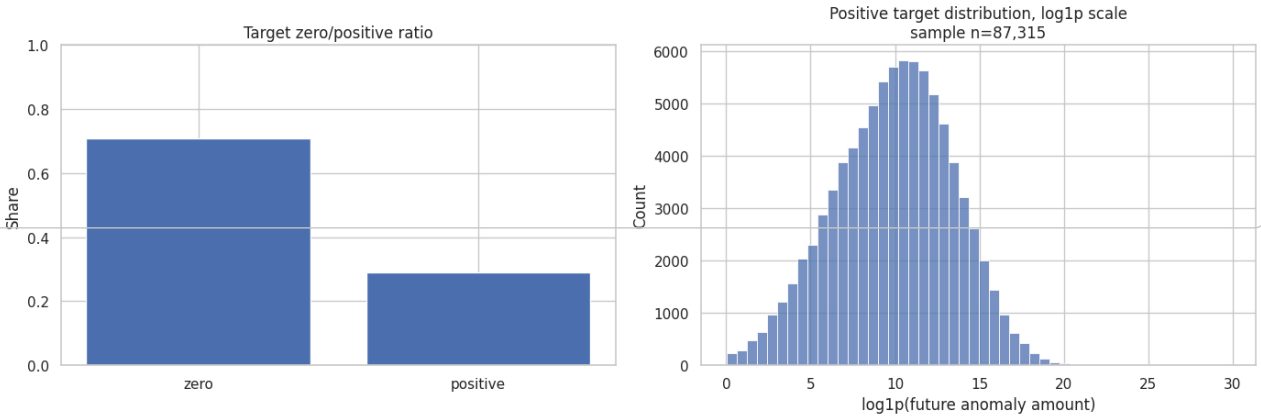



```
saved figure: figures_production/01a_chapter_coverage_heatmap.png bytes= 133005
```






```
saved figure: figures_production/01b_top100_hs_coverage_heatmap.png bytes= 178110
```




```
saved figure: figures_production/02_target_distribution.png bytes= 57939
```

	future_net_anomaly_amount
count	1.106159e+06
mean	5.379584e+08
std	5.035751e+11
min	0.000000e+00
50%	0.000000e+00
75%	3.789325e+02
90%	9.438463e+04
95%	5.527990e+05
99%	7.943168e+06
max	5.279102e+14

dtype: float64

```
# Cell 8. Build HS-month anomaly matrix for Factor + Graphical model

def build_hs_month_panel(df, agg_method=HS_AGG_METHOD):
    tmp = df.copy().dropna(subset=["rel_z", "date", "hs_code", "import_value"])
    tmp["abs_rel_z"] = tmp["rel_z"].abs()
    tmp["total_hs_month_import"] = tmp.groupby(["hs_code", "date"])["import_value"].transform("sum")
    tmp["value_weight"] = tmp["import_value"] / tmp["total_hs_month_import"]

    if agg_method == "value_weighted_abs":
        tmp["z_component"] = tmp["value_weight"] * tmp["abs_rel_z"]
    elif agg_method == "value_weighted_signed":
        tmp["z_component"] = tmp["value_weight"] * tmp["rel_z"]
    elif agg_method == "simple_abs":
        tmp["z_component"] = tmp["abs_rel_z"]
    elif agg_method == "simple_signed":
        tmp["z_component"] = tmp["rel_z"]
    else:
        raise ValueError("Invalid HS_AGG_METHOD")

    agg_func = "sum" if agg_method.startswith("value_weighted") else "mean"

    hs_month = tmp.groupby(["hs_code", "date"], as_index=False).agg(
        Z_ht=("z_component", agg_func),
        hs_import_value=("import_value", "sum"),
        n_country=("country_code", "nunique"),
        mean_abs_rel_z=("abs_rel_z", "mean"),
        max_abs_rel_z=("abs_rel_z", "max"),
    )
    return hs_month.sort_values(["hs_code", "date"]).reset_index(drop=True)


hs_month = build_hs_month_panel(crane)
print("HS-month panel:", hs_month.shape)
display(hs_month.head())

coverage = (
    hs_month.groupby("hs_code")
    .agg(
        n_month=("date", "nunique"),
        total_import=("hs_import_value", "sum"),
        mean_Z=("Z_ht", "mean"),
    )
    .reset_index()
)

eligible = (
    coverage[coverage["n_month"] >= MIN_MONTH_COVERAGE]
    .sort_values("total_import", ascending=False)
    .head(MAX_HS_NODES)
)

selected_hs = eligible["hs_code"].astype(str).tolist()
hs_month_sel = hs_month[hs_month["hs_code"].astype(str).isin(selected_hs)].copy()

print(f"Selected HS nodes: {len(selected_hs)}")
display(eligible.head(20))

Z_mat_raw = (
    hs_month_sel.pivot_table(
        index="date",
        columns="hs_code",
        values="Z_ht",
        aggfunc="mean"
    )
    .sort_index()
)

Z_mat_raw = Z_mat_raw[selected_hs]
print("Raw Z matrix shape:", Z_mat_raw.shape)
display(Z_mat_raw.head())

Z_std, Z_mu, Z_sd = safe_zscore_matrix(Z_mat_raw)

if FILL_MISSING_METHOD == "zero_after_standardize":
    Z_mat = Z_std.fillna(0.0)
elif FILL_MISSING_METHOD == "hs_mean":
    Z_mat = Z_mat_raw.apply(lambda x: x.fillna(x.mean()), axis=0)
    Z_mat, _, _ = safe_zscore_matrix(Z_mat)
    Z_mat = Z_mat.fillna(0.0)
elif FILL_MISSING_METHOD == "interpolate":
    Z_mat = Z_mat_raw.interpolate(limit_direction="both")
    Z_mat, _, _ = safe_zscore_matrix(Z_mat)
```

```
    Z_mat = Z_mat.fillna(0.0)
else:
    raise ValueError("Invalid FILL_MISSING_METHOD")

print("Final standardized Z matrix:", Z_mat.shape)

# ----------------------------
# Clean heatmap for presentation
# ----------------------------
Z_plot = Z_mat.copy()
Z_plot.index = pd.to_datetime(Z_plot.index).strftime("%Y-%m")

plt.figure(figsize=(18, 7))
sns.heatmap(
    Z_plot.T,
    cmap="vlag",
    center=0,
    vmin=-3,
    vmax=3,
    cbar_kws={"label": "standardized HS anomaly score"}
)
plt.title("HS-month standardized anomaly matrix")
plt.xlabel("Month")
plt.ylabel("HS code")
plt.xticks(rotation=90)
save_fig("figures_factor_graphical/04_hs_month_standardized_anomaly_matrix_clean.png")
```

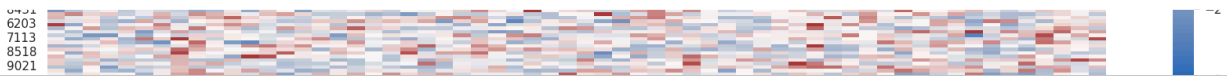
```
HS-month panel: (67372, 7)
```

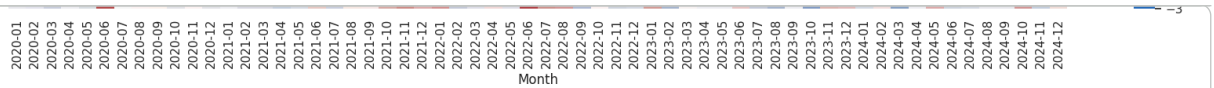
	hs_code	date	Z_ht	hs_import_value	n_country	mean_abs_rel_z	max_abs_rel_z
0	0101	2020-01-01	0.782162	787736	6	1.119444	2.331869
1	0101	2020-03-01	1.949373	4754947	4	1.109198	2.240398
2	0101	2020-04-01	1.569024	1730909	4	1.003489	1.828965
3	0101	2020-05-01	5.832146	2523463	3	10.969735	31.909205
4	0101	2020-07-01	1.274254	1334986	4	1.733045	4.549382

```
Selected HS nodes: 120
```

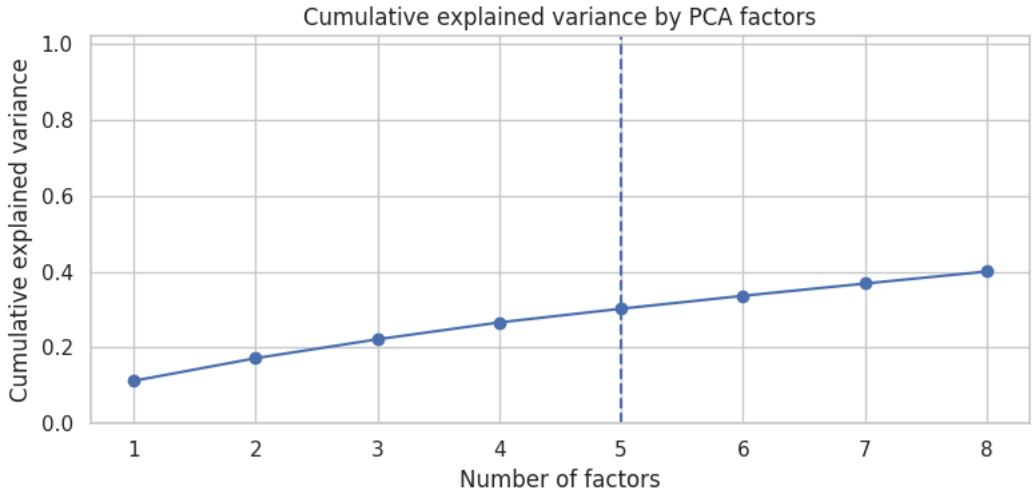
	hs_code	n_month	total_import	mean_Z
242	2709	60	388924477700	0.952461
1062	8542	60	265194879985	1.043423
244	2711	60	182381188096	0.689981
243	2710	60	110320867308	0.581579

```
# Cell 9. PCA Factor model and visualizations
max_pca = min(MAX_FACTORS_TO_PLOT, Z_mat.shape[0] - 1, Z_mat.shape[1])
pca_diag = PCA(n_components=max_pca, random_state=RANDOM_STATE)
pca_diag.fit(Z_mat.values)
explained = pca_diag.explained_variance_ratio_

plt.figure(figsize=(8, 4))
plt.plot(np.arange(1, len(explained) + 1), np.cumsum(explained), marker="o")
plt.axvline(N_FACTORS, linestyle="--")
plt.title("Cumulative explained variance by PCA factors")
plt.xlabel("Number of factors")
plt.ylabel("Cumulative explained variance")
plt.ylim(0, 1.02)
save_fig("figures_factor_graphical/05_pca_cumulative_explained_variance.png")

n_factors = min(N_FACTORS, Z_mat.shape[0] - 1, Z_mat.shape[1])
pca = PCA(n_components=n_factors, random_state=RANDOM_STATE)
F = pca.fit_transform(Z_mat.values)
loadings = pca.components_.T
Z_hat = pca.inverse_transform(F)
U = Z_mat.values - Z_hat

factor_scores = pd.DataFrame(F, index=Z_mat.index, columns=[f"F{k+1}" for k in range(n_factors)])
loading_df = pd.DataFrame(loadings, index=Z_mat.columns, columns=[f"loading_F{k+1}" for k in range(n_factors)]).reset_index().rename(colu
if "hs_code" not in loading_df.columns:
    loading_df = loading_df.rename(columns={loading_df.columns[0]: "hs_code"})
residual_mat = pd.DataFrame(U, index=Z_mat.index, columns=Z_mat.columns)

factor_scores.to_csv("output_factor_graphical/factor_scores.csv", encoding="utf-8-sig")
loading_df.to_csv("output_factor_graphical/factor_loadings.csv", index=False, encoding="utf-8-sig")
residual_mat.to_csv("output_factor_graphical/hs_idiosyncratic_residual_matrix.csv", encoding="utf-8-sig")

plt.figure(figsize=(13, 5))
for col in factor_scores.columns:
    plt.plot(factor_scores.index, factor_scores[col], marker="o", linewidth=1.7, label=col)
plt.axhline(0, linewidth=1, linestyle="--")
plt.title("Estimated common factor time series")
plt.xlabel("Month")
plt.ylabel("Factor score")
plt.legend(ncol=min(n_factors, 5))
save_fig("figures_factor_graphical/06_common_factor_time_series.png")

loading_heat = loading_df.set_index("hs_code")[[f"loading_F{k+1}" for k in range(n_factors)]]
loading_heat["dominant_factor"] = loading_heat.abs().idxmax(axis=1)
loading_heat["max_abs_loading"] = loading_heat.drop(columns="dominant_factor").abs().max(axis=1)
loading_heat_sorted = loading_heat.sort_values(["dominant_factor", "max_abs_loading"], ascending=[True, False]).drop(columns=["dominant_
plt.figure(figsize=(8, max(7, 0.14 * len(loading_heat_sorted))))
sns.heatmap(loading_heat_sorted, cmap="vlag", center=0, cbar_kws={"label": "PCA loading"})
plt.title("HS factor loading heatmap")
plt.xlabel("Common factors")
plt.ylabel("HS code")
save_fig("figures_factor_graphical/07_factor_loading_heatmap.png")

print("Factor scores:", factor_scores.shape)
print("Loadings:", loading_df.shape)
print("Residual matrix:", residual_mat.shape)
display(factor_scores.head())
display(loading_df.head())
```

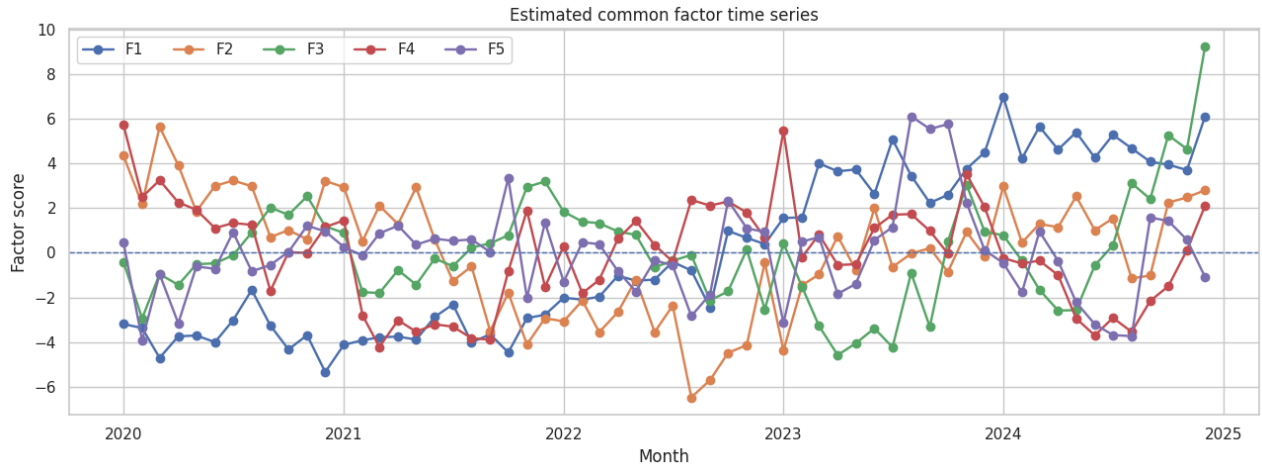




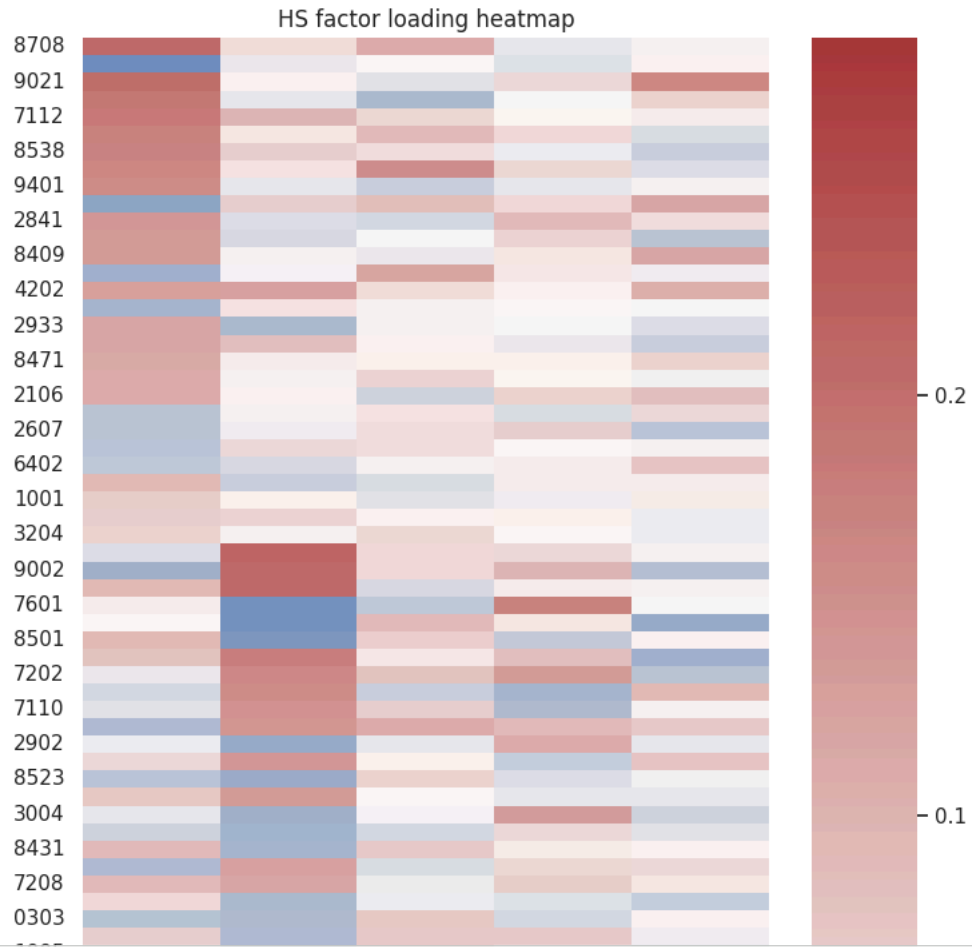

```
saved figure: figures_factor_graphical/04_hs_month_standardized_anomaly_matrix_clean.png bytes= 145140
```




```
saved figure: figures_factor_graphical/05_pca_cumulative_explained_variance.png bytes= 44336
```

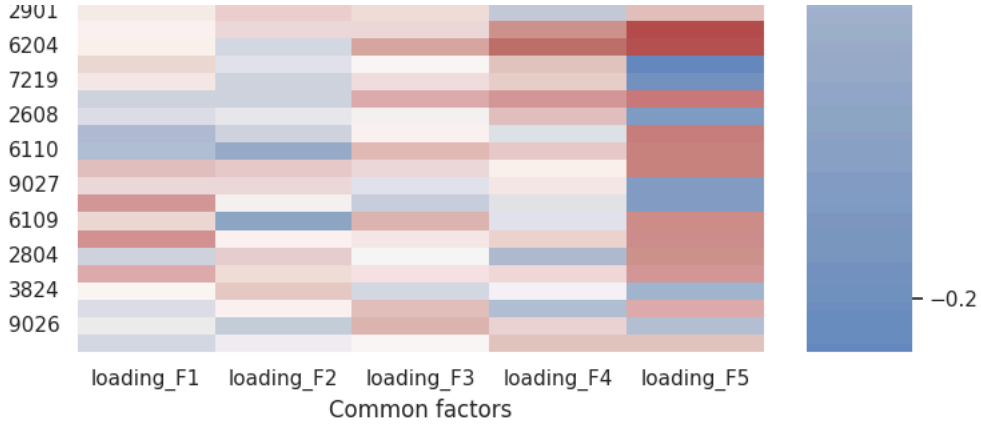



```
saved figure: figures_factor_graphical/06_common_factor_time_series.png bytes= 244290
```




```
# Cell 10. Factor VAR
F_for_var = factor_scores.copy().dropna()
if len(F_for_var) > (n_factors * VAR_MAX_LAG + 5):
```

```
    var_model = VAR(F_for_var)
    if VAR_SELECTED_LAG is None:
        lag_order_results = var_model.select_order(maxlags=min(VAR_MAX_LAG, max(1, len(F_for_var)//5)))
        selected_lag = int(lag_order_results.aic) if not np.isnan(lag_order_results.aic) else 1
        selected_lag = max(1, min(selected_lag, VAR_MAX_LAG))
    else:
        selected_lag = VAR_SELECTED_LAG
    var_res = var_model.fit(selected_lag)
    print(var_res.summary())

    var_coef_rows = []
    for lag in range(selected_lag):
        A = pd.DataFrame(var_res.coefs[lag], index=factor_scores.columns, columns=factor_scores.columns)
        A.to_csv(f"output_factor_graphical/factor_var_A_lag_{lag+1}.csv", encoding="utf-8-sig")
        plt.figure(figsize=(6, 5))
        sns.heatmap(A, cmap="vlag", center=0, annot=True, fmt=".2f", cbar_kws={"label": "VAR coefficient"})
        plt.title(f"Factor VAR coefficient matrix, lag {lag+1}")
        plt.xlabel("Source lagged factor")
        plt.ylabel("Target current factor")
        save_fig(f"figures_factor_graphical/09_factor_var_coefficients_lag_{lag+1}.png")
        for target in factor_scores.columns:
            for source in factor_scores.columns:
                var_coef_rows.append({"lag": lag+1, "source_factor": source, "target_factor": target, "coef": A.loc[target, source], "abs
    var_coef_df = pd.DataFrame(var_coef_rows)
    var_coef_df.to_csv("output_factor_graphical/factor_var_coefficients_long.csv", index=False, encoding="utf-8-sig")
else:
    var_res = None
    var_coef_df = pd.DataFrame()
    print("Not enough observations for factor VAR. Skipped.")
```




```
saved figure: figures_factor_graphical/07_factor_loading_heatmap.png bytes= 141623
Factor scores: (60, 5)
Loadings: (120, 6)
Residual matrix: (60, 120)
```

date	F1	F2	F3	F4	F5
2020-01-01	-3.192566	4.349160	-0.404427	5.729682	0.468991
2020-02-01	-3.372401	2.186011	-2.937811	2.501375	-3.931493
2020-03-01	-4.709235	5.641979	-0.952902	3.248222	-0.957349
2020-04-01	-3.744860	3.923159	-1.441549	2.258280	-3.160716
2020-05-01	-3.708062	1.848576	-0.513603	1.926260	-0.615767

	hs_code	loading_F1	loading_F2	loading_F3	loading_F4	loading_F5
0	2709	0.012249	0.012023	0.095611	0.236968	0.131628
1	8542	-0.034039	-0.057266	0.033315	0.090244	0.071899
2	2711	0.081442	0.068432	0.049275	0.014672	0.173836
3	2710	0.093231	-0.006029	-0.074917	0.209235	0.087955
4	8486	-0.083412	-0.038073	0.068210	-0.149841	-0.030854

```
  Summary of Regression Results
==================================
Model:                         VAR
Method:                        OLS
Date:           Thu, 30, Apr, 2026
Time:                     02:55:50
--------------------------------------------------------------------
No. of Equations:         5.00000    BIC:                    6.60113
Nobs:                     58.0000    HQIC:                   5.40833
Log likelihood:          -491.263    FPE:                    106.751
AIC:                      4.64726    Det(Omega_mle):         44.7990
--------------------------------------------------------------------
Results for equation F1
======================================================================
          coefficient       std. error           t-stat            prob
----------------------------------------------------------------------
const        0.255122         0.148847            1.714           0.087
L1.F1        0.556823         0.129704            4.293           0.000
L1.F2        0.106054         0.088344            1.200           0.230
L1.F3       -0.128923         0.111090           -1.161           0.246
L1.F4        0.135281         0.100777            1.342           0.179
L1.F5        0.036451         0.088529            0.412           0.681
L2.F1        0.430780         0.129879            3.317           0.001
L2.F2       -0.149010         0.091166           -1.634           0.102
L2.F3        0.188305         0.113961            1.652           0.098
L2.F4       -0.048271         0.096485           -0.500           0.617
L2.F5       -0.003704         0.085450           -0.043           0.965
======================================================================

Results for equation F2
======================================================================
          coefficient       std. error           t-stat            prob
----------------------------------------------------------------------
const        0.044774         0.218497            0.205           0.838
L1.F1       -0.044224         0.190396           -0.232           0.816
L1.F2        0.322082         0.129683            2.484           0.013
L1.F3        0.011622         0.163072            0.071           0.943
L1.F4        0.072186         0.147933            0.488           0.626
L1.F5        0.179804         0.129954            1.384           0.166
L2.F1        0.137088         0.190652            0.719           0.472
L2.F2        0.562880         0.133825            4.206           0.000
L2.F3        0.126579         0.167286            0.757           0.449
L2.F4        0.126838         0.141632            0.896           0.370
L2.F5       -0.075504         0.125434           -0.602           0.547
======================================================================

Results for equation F3
======================================================================
          coefficient       std. error           t-stat            prob
----------------------------------------------------------------------
const        0.224397         0.212917            1.054           0.292
L1.F1        0.141862         0.185534            0.765           0.445
L1.F2        0.036272         0.126371            0.287           0.774
L1.F3        0.578891         0.158908            3.643           0.000
L1.F4       -0.015168         0.144155           -0.105           0.916
L1.F5        0.131262         0.126636            1.037           0.300
L2.F1       -0.096702         0.185784           -0.521           0.603
L2.F2        0.098652         0.130407            0.756           0.449
L2.F3        0.354239         0.163014            2.173           0.030
L2.F4       -0.181225         0.138016           -1.313           0.189
L2.F5        0.011577         0.122231            0.095           0.925
======================================================================

Results for equation F4
======================================================================
          coefficient       std. error           t-stat            prob
----------------------------------------------------------------------
const       -0.031290         0.215596           -0.145           0.885
L1.F1       -0.277339         0.187868           -1.476           0.140
L1.F2        0.067561         0.127961            0.528           0.598
L1.F3        0.045095         0.160907            0.280           0.779
L1.F4        0.576988         0.145969            3.953           0.000
L1.F5        0.184779         0.128229            1.441           0.150
L2.F1        0.277424         0.188121            1.475           0.140
L2.F2       -0.191648         0.132048           -1.451           0.147
L2.F3       -0.012577         0.165065           -0.076           0.939
L2.F4        0.155873         0.139752            1.115           0.265
L2.F5       -0.105422         0.123769           -0.852           0.394
======================================================================
```

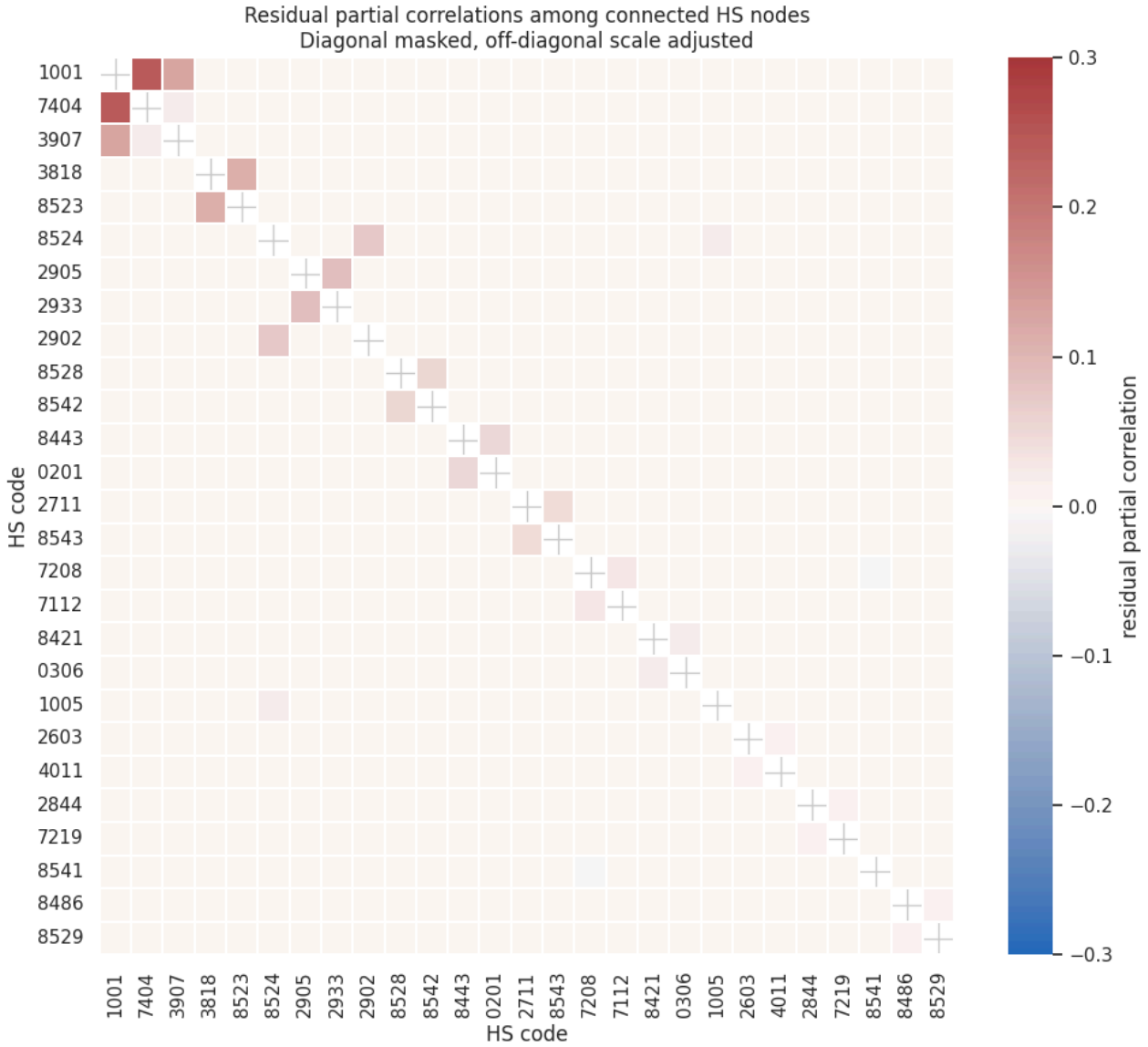
```
# Cell 11. Residual Graphical LASSO
U_std, U_mu, U_sd = safe_zscore_matrix(residual_mat)
U_std = U_std.fillna(0.0)

gl_cv = GraphicalLassoCV(alphas=8, cv=min(5, max(2, len(U_std)//12)), max_iter=1000)
```

```
gl_cv.fit(U_std.values)
precision = gl_cv.precision_
alpha_selected = gl_cv.alpha_
print("Selected Graphical LASSO alpha:", alpha_selected)

hs_codes_graph = list(U_std.columns)
precision_df = pd.DataFrame(precision, index=hs_codes_graph, columns=hs_codes_graph)
precision_df.to_csv("output_factor_graphical/residual_precision_matrix_graphical_lasso.csv", encoding="utf-8-sig")

diag = np.sqrt(np.diag(precision))
partial_corr = -precision / np.outer(diag, diag)
np.fill_diagonal(partial_corr, 1.0)
partial_corr_df = pd.DataFrame(partial_corr, index=hs_codes_graph, columns=hs_codes_graph)
partial_corr_df.to_csv("output_factor_graphical/residual_partial_correlation_matrix.csv", encoding="utf-8-sig")

edge_rows = []
p = len(hs_codes_graph)
for i in range(p):
    for j in range(i + 1, p):
        pc = partial_corr[i, j]
        om = precision[i, j]
        if abs(pc) > 1e-12:
            edge_rows.append({"hs_i": hs_codes_graph[i], "hs_j": hs_codes_graph[j], "omega_ij": om, "partial_corr": pc, "abs_partial_cc
edge_df = pd.DataFrame(edge_rows).sort_values("abs_partial_corr", ascending=False)
edge_df.to_csv("output_factor_graphical/residual_graphical_lasso_edges_all.csv", index=False, encoding="utf-8-sig")
print("Edges with nonzero partial corr:", len(edge_df))
display(edge_df.head(20))

top_connected = (
    edge_df.assign(weight=edge_df["abs_partial_corr"])
    .melt(id_vars=["weight"], value_vars=["hs_i", "hs_j"], value_name="hs_code")
    .groupby("hs_code")["weight"].sum().sort_values(ascending=False).head(min(50, len(hs_codes_graph))).index.tolist()
)


heat_pc = partial_corr_df.loc[top_connected, top_connected].copy()

mask = np.eye(len(heat_pc), dtype=bool)

plt.figure(figsize=(10, 9))
sns.heatmap(
    heat_pc,
    mask=mask,
    cmap="vlag",
    center=0,
    vmin=-0.30,
    vmax=0.30,
    linewidths=0.2,
    linecolor="white",
    cbar_kws={"label": "residual partial correlation"}
)

plt.title(
    "Residual partial correlations among connected HS nodes₩n"
    "Diagonal masked, off-diagonal scale adjusted"
)
plt.xlabel("HS code")
plt.ylabel("HS code")
plt.tight_layout()
plt.savefig(
    "figures_factor_graphical/11_residual_partial_correlation_heatmap_masked.png",
    dpi=160,
    bbox_inches="tight"
)
plt.show()
```

```
saved figure: figures_factor_graphical/09_factor_var_coefficients_lag_2.png bytes= 67509
```

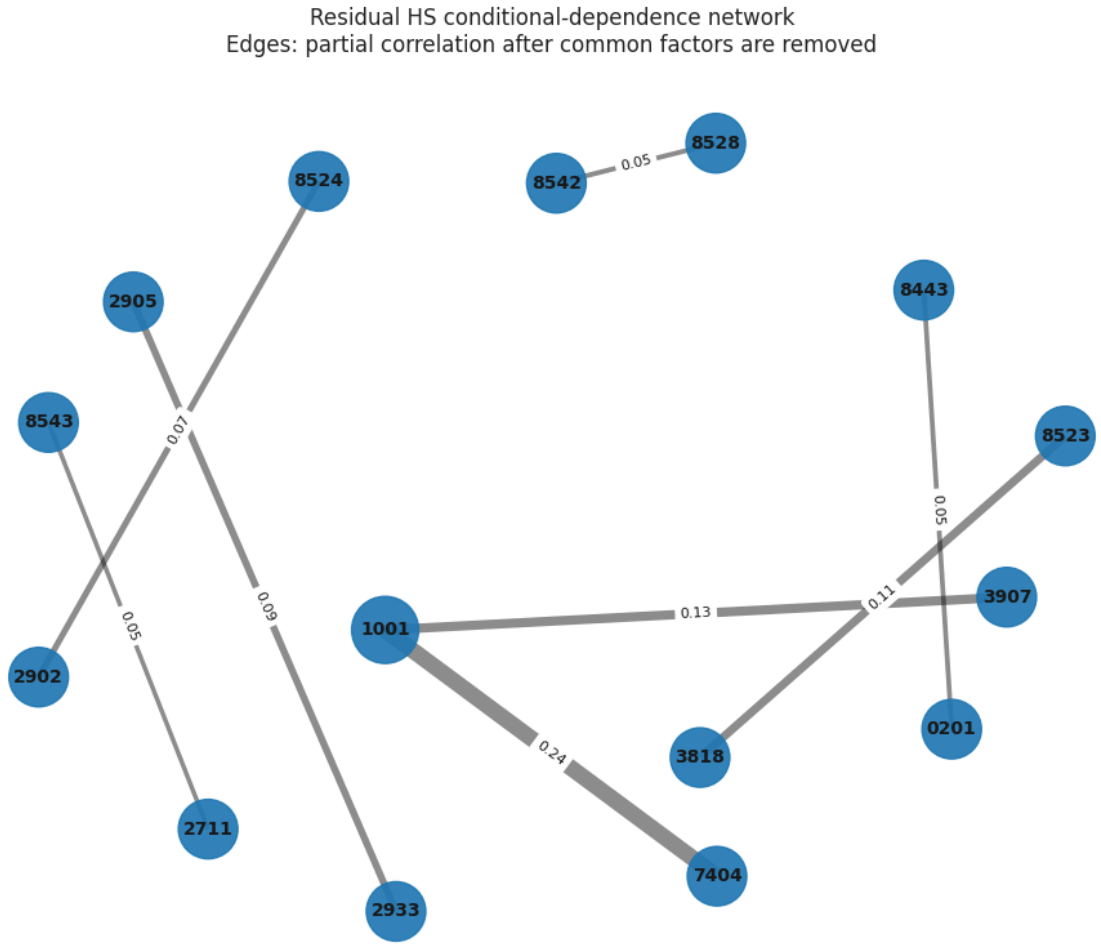
```
Selected Graphical LASSO alpha: 0.41828551879207737
Edges with nonzero partial corr: 16
```

	hs_i	hs_j	omega_ij	partial_corr	abs_partial_corr	sign
8	7404	1001	-0.267144	0.244420	0.244420	positive
14	1001	3907	-0.135849	0.127141	0.127141	positive
7	8523	3818	-0.114120	0.110839	0.110839	positive
12	2905	2933	-0.090932	0.088713	0.088713	positive
11	8524	2902	-0.074900	0.073243	0.073243	positive
0	8542	8528	-0.053381	0.052347	0.052347	positive
15	8443	0201	-0.051169	0.050189	0.050189	positive
1	2711	8543	-0.045979	0.045121	0.045121	positive
6	7208	7112	-0.027187	0.026714	0.026714	positive
10	8421	0306	-0.020272	0.019926	0.019926	positive
5	1005	8524	-0.019375	0.018994	0.018994	positive
9	7404	3907	-0.017875	0.016864	0.016864	positive
4	2603	4011	-0.013784	0.013551	0.013551	positive
13	7219	2844	-0.010848	0.010666	0.010666	positive
3	8541	7208	0.008980	-0.008827	0.008827	negative
2	8486	8529	-0.006151	0.006049	0.006049	positive




```
edge_plot = edge_df[edge_df["abs_partial_corr"] >= 0.03].copy()

G_resid = nx.Graph()
```

```
for _, r in edge_plot.iterrows():
    G_resid.add_edge(
        r["hs_i"],
        r["hs_j"],
        weight=r["abs_partial_corr"],
        partial_corr=r["partial_corr"],
        sign=r["sign"]
    )

plt.figure(figsize=(10, 8))
pos = nx.spring_layout(G_resid, seed=123, k=0.8)

weights = [G_resid[u][v]["weight"] for u, v in G_resid.edges()]
max_w = max(weights) if weights else 1

edge_widths = [1 + 8 * w / max_w for w in weights]

node_sizes = [
    800 + 300 * G_resid.degree(n)
    for n in G_resid.nodes()
]

nx.draw_networkx_nodes(
    G_resid,
    pos,
    node_size=node_sizes,
    alpha=0.9
)

nx.draw_networkx_edges(
    G_resid,
    pos,
    width=edge_widths,
    alpha=0.5
)

nx.draw_networkx_labels(
    G_resid,
    pos,
    font_size=10,
    font_weight="bold"
)

edge_labels = {
    (u, v): f"{G_resid[u][v]['partial_corr']:.2f}"
    for u, v in G_resid.edges()
}

nx.draw_networkx_edge_labels(
    G_resid,
    pos,
    edge_labels=edge_labels,
    font_size=8
)

plt.title("Residual HS conditional-dependence network₩nEdges: partial correlation after common factors are removed")
plt.axis("off")
plt.tight_layout()
plt.savefig(
    "figures_factor_graphical/13_residual_hs_network_clean.png",
    dpi=160,
    bbox_inches="tight"
)
plt.show()
```




```
# Cell 12. Optional bootstrap edge stability

def block_bootstrap_indices(T, block_size, rng):
    starts = np.arange(0, T - block_size + 1)
    idx = []
    while len(idx) < T:
        s = rng.choice(starts)
        idx.extend(range(s, s + block_size))
    return np.array(idx[:T])

if RUN_BOOTSTRAP_STABILITY:
    rng = np.random.default_rng(RANDOM_STATE)
    T, p = U_std.shape
    edge_count = np.zeros((p, p), dtype=float)
    for b in range(BOOTSTRAP_B):
        idx = block_bootstrap_indices(T, BOOTSTRAP_BLOCK_SIZE, rng)
        Ub = U_std.values[idx, :]
        try:
            gl_b = GraphicalLasso(alpha=alpha_selected, max_iter=1000)
            gl_b.fit(Ub)
            prec_b = gl_b.precision_
            diag_b = np.sqrt(np.diag(prec_b))
            pc_b = -prec_b / np.outer(diag_b, diag_b)
            np.fill_diagonal(pc_b, 0.0)
            selected_b = np.abs(pc_b) >= EDGE_ABS_PARTIAL_CORR_THRESHOLD
            edge_count += selected_b.astype(float)
        except Exception as err:
            print(f"Bootstrap {b} failed: {err}")
    stability = edge_count / BOOTSTRAP_B
    np.fill_diagonal(stability, 0.0)
    stability_df = pd.DataFrame(stability, index=hs_codes_graph, columns=hs_codes_graph)
    stability_df.to_csv("output_factor_graphical/residual_edge_stability_matrix.csv", encoding="utf-8-sig")
    stability_rows = []
    for i in range(p):
        for j in range(i + 1, p):
            stability_rows.append({"hs_i": hs_codes_graph[i], "hs_j": hs_codes_graph[j], "edge_stability": stability[i, j], "partial_co
    stability_edge_df = pd.DataFrame(stability_rows).sort_values(["edge_stability", "abs_partial_corr"], ascending=False)
    stability_edge_df.to_csv("output_factor_graphical/residual_edge_stability_edges.csv", index=False, encoding="utf-8-sig")
    print("Bootstrap stability complete.")
```

```
    display(stability_edge_df.head(20))
else:
    stability_df = None
    stability_edge_df = edge_df.copy()
    stability_edge_df["edge_stability"] = np.nan
    print("Bootstrap stability skipped.")
```

```
Bootstrap stability skipped.
```

```
# Cell 13. Build HS conditional-dependence network and graph features
if RUN_BOOTSTRAP_STABILITY:
    final_edges = stability_edge_df[(stability_edge_df["edge_stability"] >= EDGE_STABILITY_THRESHOLD) & (stability_edge_df["abs_partial_c
else:
    final_edges = edge_df[edge_df["abs_partial_corr"] >= EDGE_ABS_PARTIAL_CORR_THRESHOLD].copy()

if len(final_edges) > TOP_EDGES_TO_PLOT:
    sort_cols = ["edge_stability", "abs_partial_corr"] if "edge_stability" in final_edges.columns else ["abs_partial_corr"]
    final_edges_plot = final_edges.sort_values(sort_cols, ascending=False).head(TOP_EDGES_TO_PLOT).copy()
else:
    final_edges_plot = final_edges.copy()

G = nx.Graph()
for hs in hs_codes_graph:
    G.add_node(hs)
for _, r in final_edges_plot.iterrows():
    G.add_edge(r["hs_i"], r["hs_j"], weight=float(r["abs_partial_corr"]), partial_corr=float(r["partial_corr"]), stability=float(r.get("e

node_info = hs_month_sel.groupby("hs_code").agg(mean_Z=("Z_ht", "mean"), total_import=("hs_import_value", "sum"), n_month=("date", "nunic
for n in G.nodes():
    G.nodes[n]["mean_Z"] = node_info.get(n, {}).get("mean_Z", 0.0)
    G.nodes[n]["total_import"] = node_info.get(n, {}).get("total_import", 0.0)

weighted_degree = dict(G.degree(weight="weight"))
betweenness = nx.betweenness_centrality(G, weight="weight") if G.number_of_edges() > 0 else {n: 0 for n in G.nodes()}
try:
    eigenvector = nx.eigenvector_centrality_numpy(G, weight="weight") if G.number_of_edges() > 0 else {n: 0 for n in G.nodes()}
except Exception:
    eigenvector = {n: 0 for n in G.nodes()}

centrality_df = pd.DataFrame({
    "hs_code": list(G.nodes()),
    "weighted_degree": [weighted_degree.get(n, 0.0) for n in G.nodes()],
    "betweenness": [betweenness.get(n, 0.0) for n in G.nodes()],
    "eigenvector": [eigenvector.get(n, 0.0) for n in G.nodes()],
    "mean_Z": [G.nodes[n].get("mean_Z", 0.0) for n in G.nodes()],
    "total_import": [G.nodes[n].get("total_import", 0.0) for n in G.nodes()],
}).sort_values("weighted_degree", ascending=False)
centrality_df.to_csv("output_factor_graphical/hs_graph_centrality.csv", index=False, encoding="utf-8-sig")
display(centrality_df.head(20))

G_plot = G.copy()
G_plot.remove_nodes_from(list(nx.isolates(G_plot)))
plt.figure(figsize=(14, 11))
if G_plot.number_of_edges() == 0:
    plt.text(0.5, 0.5, "No edges under current thresholds", ha="center", va="center", fontsize=15)
    plt.axis("off")
else:
    pos = nx.spring_layout(G_plot, seed=RANDOM_STATE, k=0.8)
    deg_plot = dict(G_plot.degree(weight="weight"))
    node_sizes = [350 + 3500 * deg_plot.get(n, 0) / max(1e-9, max(deg_plot.values())) for n in G_plot.nodes()]
    node_colors = [node_info.get(n, {}).get("mean_Z", 0.0) for n in G_plot.nodes()]
    max_w = max([G_plot[a][b]["weight"] for a, b in G_plot.edges()])
    edge_widths = [0.6 + 5.0 * G_plot[u][v]["weight"] / max(1e-9, max_w) for u, v in G_plot.edges()]
    nx.draw_networkx_edges(G_plot, pos, width=edge_widths, alpha=0.35)
    nodes = nx.draw_networkx_nodes(G_plot, pos, node_size=node_sizes, node_color=node_colors, cmap="viridis", alpha=0.90)
    nx.draw_networkx_labels(G_plot, pos, font_size=9, font_weight="bold")
    plt.colorbar(nodes, label="Mean HS anomaly score")
    plt.title("Factor-adjusted HS conditional-dependence network")
    plt.axis("off")
save_fig("figures_factor_graphical/13_factor_adjusted_hs_network.png")

# Graph feature construction
W = np.zeros((len(hs_codes_graph), len(hs_codes_graph)))
hs_to_idx = {h: i for i, h in enumerate(hs_codes_graph)}
for u, v, d in G.edges(data=True):
    i, j = hs_to_idx[u], hs_to_idx[v]
    W[i, j] = d["weight"]
    W[j, i] = d["weight"]
row_sum = W.sum(axis=1, keepdims=True)
W_norm = np.divide(W, row_sum, out=np.zeros_like(W), where=row_sum > 0)
U_abs = np.abs(residual_mat[hs_codes_graph].values)
neighbor_risk = U_abs @ W_norm.T
```

```
neighbor_risk_df = pd.DataFrame(neighbor_risk, index=residual_mat.index, columns=hs_codes_graph)
own_resid_abs_df = pd.DataFrame(U_abs, index=residual_mat.index, columns=hs_codes_graph)
loading_matrix = loading_df.set_index("hs_code").loc[hs_codes_graph, [f"loading_F{k+1}" for k in range(n_factors)]].values
factor_exposure = np.abs(factor_scores.values) @ np.abs(loading_matrix).T
factor_exposure_df = pd.DataFrame(factor_exposure, index=factor_scores.index, columns=hs_codes_graph)

def wide_to_long_feature(wide_df, value_name):
    wide_df = wide_df.copy()
    wide_df.index.name = "date"
    return wide_df.reset_index().melt(id_vars="date", var_name="hs_code", value_name=value_name)

graph_feature_hs_month = wide_to_long_feature(neighbor_risk_df, "graph_neighbor_risk")
graph_feature_hs_month = graph_feature_hs_month.merge(wide_to_long_feature(own_resid_abs_df, "graph_own_residual_abs"), on=["date", "hs_c
graph_feature_hs_month = graph_feature_hs_month.merge(wide_to_long_feature(factor_exposure_df, "graph_factor_exposure"), on=["date", "hs_
graph_feature_hs_month = graph_feature_hs_month.merge(
    centrality_df[["hs_code", "weighted_degree", "betweenness", "eigenvector"]].rename(columns={"weighted_degree": "graph_weighted_degree
    on="hs_code", how="left"
)
factor_scores_long = factor_scores.reset_index().rename(columns={factor_scores.index.name or "index": "date"})
graph_feature_hs_month = graph_feature_hs_month.merge(factor_scores_long, on="date", how="left")
graph_feature_hs_month.to_csv("output_factor_graphical/graph_features_hs_month.csv", index=False, encoding="utf-8-sig")

crane_graph = crane.merge(graph_feature_hs_month, on=["date", "hs_code"], how="left")
for col in ["graph_neighbor_risk", "graph_own_residual_abs", "graph_factor_exposure", "graph_weighted_degree", "graph_betweenness", "gra
    if col in crane_graph.columns:
        crane_graph[col] = crane_graph[col].fillna(0.0)
for col in factor_scores.columns:
    if col in crane_graph.columns:
        crane_graph[col] = crane_graph[col].fillna(0.0)
crane_graph.to_csv("output_factor_graphical/crane_cell_features_with_graph_features.csv", index=False, encoding="utf-8-sig")
print("Cell-level data with graph features:", crane_graph.shape)
display(crane_graph.head())
```

```
# Cell 14. Downstream CRANE baseline vs graph-augmented Hurdle model

def top_k_metrics(df, score_col, label_col, amount_col, k_values=K_VALUES):
    tmp = df.copy().sort_values(score_col, ascending=False)
    out = {}
    base_rate = tmp[label_col].mean()
    total_pos = tmp[label_col].sum()
    total_amount = tmp[amount_col].sum()
    for k in k_values:
        top = tmp.head(min(k, len(tmp)))
        out[f"precision_at_{k}"] = top[label_col].mean()
        out[f"recall_at_{k}"] = np.nan if total_pos == 0 else top[label_col].sum() / total_pos
        out[f"lift_at_{k}"] = np.nan if base_rate == 0 else top[label_col].mean() / base_rate
        out[f"captured_amount_at_{k}"] = top[amount_col].sum()
        out[f"captured_amount_share_at_{k}"] = np.nan if total_amount == 0 else top[amount_col].sum() / total_amount
    return out


def evaluate_score(df, score, name, label_col="material_anomaly", amount_col="future_net_anomaly_amount", amount_like=False):
    tmp = df.copy()
    tmp["score_eval"] = np.asarray(score, dtype=float)
    tmp["score_eval"] = tmp["score_eval"].replace([np.inf, -np.inf], np.nan)
    tmp["score_eval"] = tmp["score_eval"].fillna(tmp["score_eval"].median())
    y_label = tmp[label_col].astype(int).to_numpy()
    s = tmp["score_eval"].to_numpy()
    y_amount = tmp[amount_col].to_numpy(dtype=float)
    rec = {
        "model": name,
        "n": len(tmp),
        "base_rate": tmp[label_col].mean(),
        "auc": roc_auc_score(y_label, s) if len(np.unique(y_label)) > 1 else np.nan,
        "average_precision": average_precision_score(y_label, s) if len(np.unique(y_label)) > 1 else np.nan,
    }
    if amount_like:
        pred = np.maximum(s, 0)
        rec["rmse"] = np.sqrt(mean_squared_error(y_amount, pred))
        rec["mae"] = mean_absolute_error(y_amount, pred)
        rec["r2"] = r2_score(y_amount, pred)
    else:
        rec["rmse"] = np.nan
        rec["mae"] = np.nan
        rec["r2"] = np.nan
    tmp["score_eval"] = s
    rec.update(top_k_metrics(tmp, "score_eval", label_col, amount_col))
    return rec
```

```
if RUN_DOWNSTREAM_LIGHTGBM:
    target_col = "future_net_anomaly_amount"
    model_df = crane_graph.replace([np.inf, -np.inf], np.nan).copy()
    zero_fill_features = [
        "rank_shift", "abs_rank_shift", "price_return", "price_return_roll3", "price_volatility_roll3",
        "import_growth", "import_growth_roll3", "weight_growth", "ht_median_return", "ht_mad_return",
        "total_import_return", "price_distribution_shift", "anomaly_abs_roll3", "anomaly_up_roll3",
        "anomaly_down_roll3", "abs_rel_z_roll3",
    ]
    for col in zero_fill_features:
        if col in model_df.columns:
            model_df[col] = model_df[col].fillna(0.0)

    base_features = [
        "rel_z", "abs_rel_z", "counterfactual_dev", "counterfactual_dev_scaled", "price_rank_pct",
        "rank_shift", "abs_rank_shift", "network_isolation_score", "idiosyncratic_current_score",
        "anomaly_abs_roll3", "anomaly_up_roll3", "anomaly_down_roll3", "abs_rel_z_roll3",
        "price_return", "price_return_roll3", "price_volatility_roll3", "import_growth", "import_growth_roll3",
        "weight_growth", "import_share_country", "hhi_hs_month", "concentration_risk", "sync_score",
        "ht_median_return", "ht_mad_return", "total_import_return", "price_distribution_shift",
        "log_price", "log_import_value", "log_import_weight", "ht_country_count", "month_sin", "month_cos",
    ]
    base_features = [c for c in base_features if c in model_df.columns]
    graph_features = [
        "graph_neighbor_risk", "graph_own_residual_abs", "graph_factor_exposure", "graph_weighted_degree", "graph_betweenness", "graph_e
    ] + [c for c in factor_scores.columns if c in model_df.columns]
    graph_features = [c for c in graph_features if c in model_df.columns]
    leakage_prefixes = ("next_", "future_")
    base_features = [c for c in base_features if not c.startswith(leakage_prefixes)]
    graph_features = [c for c in graph_features if not c.startswith(leakage_prefixes)]

    model_df = model_df.dropna(subset=base_features + graph_features + [target_col, "date"]).copy()
    train_df = model_df[model_df["date"] < pd.to_datetime(TRAIN_END)].copy()
    valid_df = model_df[(model_df["date"] >= pd.to_datetime(TRAIN_END)) & (model_df["date"] < pd.to_datetime(VALID_END))].copy()
    test_df = model_df[model_df["date"] >= pd.to_datetime(VALID_END)].copy()

    if train_df.empty or valid_df.empty or test_df.empty:
        dates = sorted(model_df["date"].dropna().unique())
        valid_cut = dates[int(len(dates) * 0.70)]
        test_cut = dates[int(len(dates) * 0.85)]
        train_df = model_df[model_df["date"] < valid_cut].copy()
        valid_df = model_df[(model_df["date"] >= valid_cut) & (model_df["date"] < test_cut)].copy()
        test_df = model_df[model_df["date"] >= test_cut].copy()
        print("Auto temporal split used:", valid_cut, test_cut)

    material_threshold = float(train_df[target_col].quantile(MATERIAL_QUANTILE))
    if material_threshold <= 0 and (train_df[target_col] > 0).any():
        material_threshold = float(train_df.loc[train_df[target_col] > 0, target_col].median())
    for df_ in [train_df, valid_df, test_df]:
        df_["material_anomaly"] = (df_[target_col] >= material_threshold).astype(int)

    print("Base features:", len(base_features))
    print("Graph features:", len(graph_features))
    print("Material threshold:", material_threshold)
    for name, df_ in [("train", train_df), ("valid", valid_df), ("test", test_df)]:
        print(name, df_.shape, df_["date"].min(), df_["date"].max(), "material rate", df_["material_anomaly"].mean())
else:
    print("Downstream LightGBM skipped.")
```

```
Base features: 33
Graph features: 11
Material threshold: 89404.67279628909
train (658731, 77) 2020-02-01 00:00:00 2022-12-01 00:00:00 material rate 0.10000136626331538
valid (222399, 77) 2023-01-01 00:00:00 2023-12-01 00:00:00 material rate 0.10531522174110496
test (205031, 77) 2024-01-01 00:00:00 2024-11-01 00:00:00 material rate 0.1038672200789149
```

```
# Cell 15. Fit downstream Hurdle models and compare

def fit_hurdle_lgbm(train_df, valid_df, test_df, features, model_name):
    X_train = train_df[features]
    X_valid = valid_df[features]
    X_test = test_df[features]
    y_train_cls = train_df["material_anomaly"].to_numpy(dtype=int)
    y_valid_cls = valid_df["material_anomaly"].to_numpy(dtype=int)

    pos = max(y_train_cls.sum(), 1)
    neg = max(len(y_train_cls) - y_train_cls.sum(), 1)
    clf = lgb.LGBMClassifier(
        objective="binary", num_leaves=31, min_child_samples=40, learning_rate=0.03,
        n_estimators=2000, subsample=0.85, colsample_bytree=0.85,
        scale_pos_weight=neg/pos, random_state=RANDOM_STATE, n_jobs=-1, verbosity=-1,
    )
```

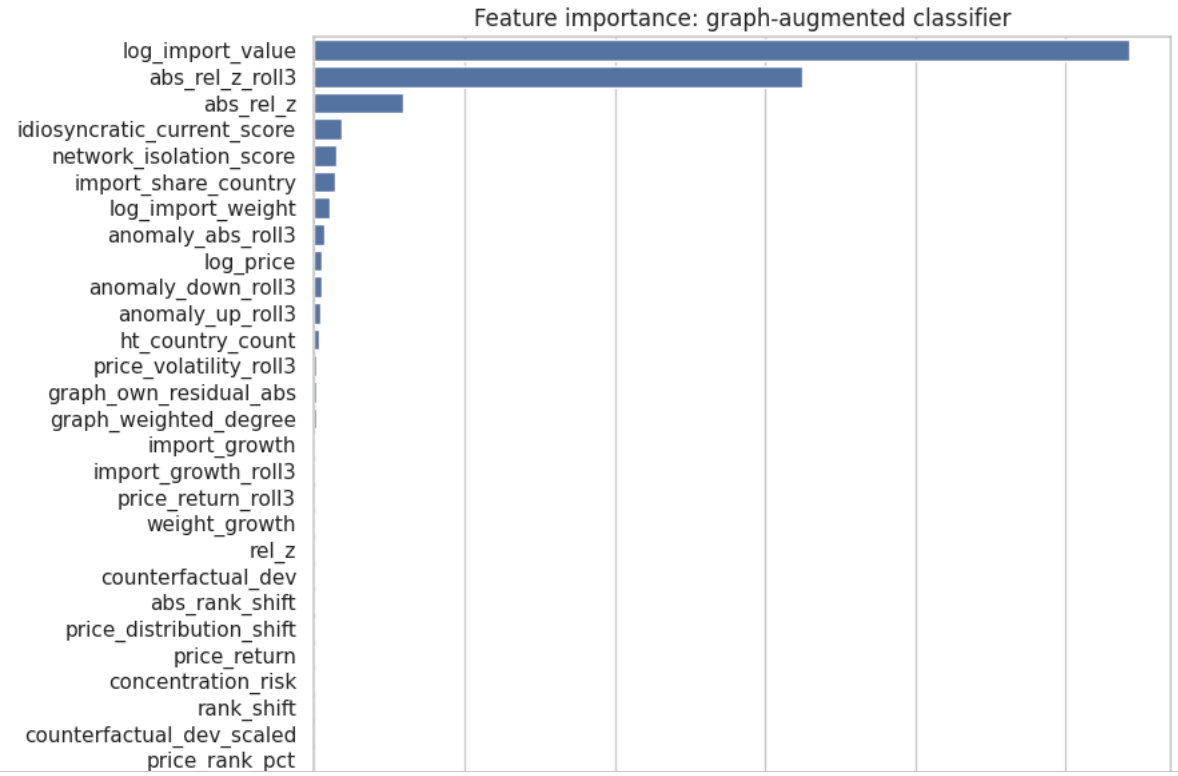
```
    clf.fit(X_train, y_train_cls, eval_set=[(X_valid, y_valid_cls)], eval_metric="auc", callbacks=[lgb.early_stopping(stopping_rounds=80
    valid_prob = clf.predict_proba(X_valid, num_iteration=clf.best_iteration_)[:, 1]
    test_prob = clf.predict_proba(X_test, num_iteration=clf.best_iteration_)[:, 1]

    sev_train = train_df[train_df["material_anomaly"] == 1].copy()
    sev_valid = valid_df[valid_df["material_anomaly"] == 1].copy()
    if len(sev_train) < 50:
        sev_train = train_df[train_df[target_col] > 0].copy()
    if len(sev_valid) < 30:
        sev_valid = valid_df[valid_df[target_col] > 0].copy()

    if len(sev_train) < 20 or len(sev_valid) < 10:
        print(f"{model_name}: Too few positive samples for severity model. Using median fallback.")
        fallback = max(float(sev_train[target_col].median()), EPS) if len(sev_train) else max(float(train_df[target_col].median()), EPS)
        valid_sev = np.repeat(fallback, len(valid_df))
        test_sev = np.repeat(fallback, len(test_df))
        sev = None
    else:
        sev = lgb.LGBMRegressor(
            objective="tweedie", metric="tweedie", tweedie_variance_power=1.5,
            num_leaves=31, min_child_samples=15, learning_rate=0.03,
            n_estimators=2000, subsample=0.90, colsample_bytree=0.90,
            random_state=RANDOM_STATE, n_jobs=-1, verbosity=-1,
        )
        sev.fit(sev_train[features], sev_train[target_col], eval_set=[(sev_valid[features], sev_valid[target_col])], eval_metric="tweedie
        valid_sev = np.maximum(sev.predict(X_valid, num_iteration=sev.best_iteration_), 0)
        test_sev = np.maximum(sev.predict(X_test, num_iteration=sev.best_iteration_), 0)

    valid_raw = valid_prob * valid_sev
    test_raw = test_prob * test_sev
    scale = valid_df[target_col].sum() / valid_raw.sum() if valid_raw.sum() > 0 else 1.0
    scale = float(np.clip(scale, 0.05, 20.0))
    valid_amount = valid_raw * scale
    test_amount = test_raw * scale

    valid_out = valid_df.copy()
    test_out = test_df.copy()
    for out, prob, sev_pred, amount in [(valid_out, valid_prob, valid_sev, valid_amount), (test_out, test_prob, test_sev, test_amount)]:
        out[f"{model_name}_prob"] = prob
        out[f"{model_name}_severity"] = sev_pred
        out[f"{model_name}_hurdle_amount"] = amount
        out[f"{model_name}_rank_score"] = pct_rank_score(amount) * 0.6 + pct_rank_score(prob) * 0.4

    eval_rows = [
        evaluate_score(test_out, test_out[f"{model_name}_prob"], f"{model_name}: probability", amount_like=False),
        evaluate_score(test_out, test_out[f"{model_name}_hurdle_amount"], f"{model_name}: hurdle expected amount", amount_like=True),
        evaluate_score(test_out, test_out[f"{model_name}_rank_score"], f"{model_name}: hybrid rank score", amount_like=False),
    ]

    clf_imp = pd.DataFrame({"feature": features, "gain": clf.booster_.feature_importance(importance_type="gain"), "split": clf.booster_.
    if sev is not None:
        sev_imp = pd.DataFrame({"feature": features, "gain": sev.booster_.feature_importance(importance_type="gain"), "split": sev.booste
    else:
        sev_imp = pd.DataFrame(columns=["feature", "gain", "split", "model"])
    return {"clf": clf, "sev": sev, "valid": valid_out, "test": test_out, "eval": pd.DataFrame(eval_rows), "feature_importance": pd.conca

if RUN_DOWNSTREAM_LIGHTGBM:
    baseline_result = fit_hurdle_lgbm(train_df, valid_df, test_df, base_features, "baseline_crane")
    graph_result = fit_hurdle_lgbm(train_df, valid_df, test_df, base_features + graph_features, "graph_augmented")

    model_comparison = pd.concat([baseline_result["eval"], graph_result["eval"]], ignore_index=True)
    if "import_value" in test_df.columns:
        model_comparison = pd.concat([pd.DataFrame([evaluate_score(test_df, test_df["import_value"], "naive: import value", amount_like=
    if "abs_rel_z" in test_df.columns:
        current_excess = test_df["import_value"] * np.maximum(test_df["abs_rel_z"] - ANOMALY_K, 0)
        model_comparison = pd.concat([pd.DataFrame([evaluate_score(test_df, current_excess, "naive: current excess amount", amount_like=

    model_comparison = model_comparison.sort_values(["average_precision", "precision_at_30"], ascending=False)
    model_comparison.to_csv("output_factor_graphical/downstream_model_comparison.csv", index=False, encoding="utf-8-sig")
    feature_importance_all = pd.concat([baseline_result["feature_importance"], graph_result["feature_importance"]], ignore_index=True)
    feature_importance_all.to_csv("output_factor_graphical/downstream_feature_importance.csv", index=False, encoding="utf-8-sig")
    display(model_comparison)

    metric_cols = ["average_precision", "precision_at_30", "precision_at_50", "captured_amount_share_at_30", "captured_amount_share_at_5
    plot_df = model_comparison[["model"] + [c for c in metric_cols if c in model_comparison.columns]].melt(id_vars="model", var_name="me
    plt.figure(figsize=(13, 7))
    sns.barplot(data=plot_df, y="model", x="value", hue="metric")
    plt.title("Downstream early-warning performance comparison")
    plt.xlabel("Metric value")
    plt.ylabel("")
    save_fig("figures_factor_graphical/16_downstream_performance_comparison.png")

    imp_graph_clf = feature_importance_all[feature_importance_all["model"] == "graph_augmented_classifier"].copy()
```

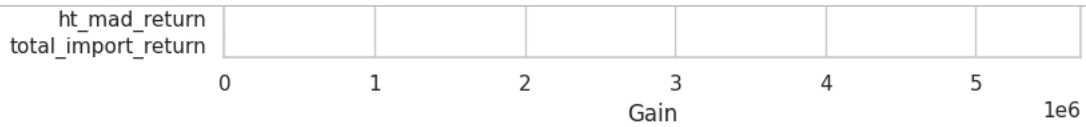
```
    plt.figure(figsize=(9, 7))
    sns.barplot(data=imp_graph_clf.head(30), y="feature", x="gain")
    plt.title("Feature importance: graph-augmented classifier")
    plt.xlabel("Gain")
    plt.ylabel("")
    save_fig("figures_factor_graphical/17_feature_importance_graph_augmented_classifier.png")
else:
    print("Prediction fitting skipped.")
```

	model	n	base_rate	auc	average_precision	rmse	mae	r2	precision_at_10
2	baseline_crane: probability	205031	0.103867	0.943676	0.711798	NaN	NaN	NaN	1.0
5	graph_augmented: probability	205031	0.103867	0.943773	0.711413	NaN	NaN	NaN	0.9
7	graph_augmented: hybrid rank score	205031	0.103867	0.929061	0.642416	NaN	NaN	NaN	1.0
4	baseline_crane: hybrid rank score	205031	0.103867	0.928800	0.640525	NaN	NaN	NaN	1.0
0	naive: current excess amount	205031	0.103867	0.794048	0.533344	1.130178e+07	5.615421e+05	0.120703	0.8
6	graph_augmented: hurdle expected	205031	0.103867	0.888593	0.422545	1.017251e+07	6.665444e+05	0.287643	0.6

```
# Cell 16. Export final warning list and download outputs if needed
if RUN_DOWNSTREAM_LIGHTGBM:
    final_warning = graph_result["test"].copy().sort_values(["graph_augmented_rank_score", "graph_augmented_hurdle_amount"], ascending=
    final_warning["warning_rank"] = np.arange(1, len(final_warning) + 1)
    warning_cols = [
        "warning_rank", "date", "next_date", "hs_code", "hs_name", "country_code", "country_name",
        "import_value", "import_weight", "unit_price", "rel_z", "next_rel_z", "future_net_anomaly_amount", "material_anomaly",
        "graph_augmented_prob", "graph_augmented_hurdle_amount", "graph_augmented_rank_score",
        "graph_neighbor_risk", "graph_own_residual_abs", "graph_factor_exposure", "graph_weighted_degree", "graph_betweenness", "graph_
    ]
    warning_cols = [c for c in warning_cols if c in final_warning.columns]
    final_warning[warning_cols].to_csv("output_factor_graphical/final_warning_list_graph_augmented.csv", index=False, encoding="utf-8-s
    display(final_warning[warning_cols].head(50))

print("Saved core files:")
for p in [
    "data_raw/customs_raw_production.csv",
    "data_processed/trade_clean_production.csv",
    "output_production/crane_features_production.csv",
    "output_factor_graphical/crane_cell_features_with_graph_features.csv",
    "output_factor_graphical/downstream_model_comparison.csv",
    "output_factor_graphical/final_warning_list_graph_augmented.csv",
]:
    path = Path(p)
    print(p, "exists=", path.exists(), "size_MB=", round(path.stat().st_size/1024/1024, 2) if path.exists() else None)

# Colab에서 파일 다운로드가 필요하면 아래 주석을 해제하세요.
# from google.colab import files
# files.download("output_production/crane_features_production.csv")
# files.download("output_factor_graphical/downstream_model_comparison.csv")
```

```
saved figure: figures_factor_graphical/16_downstream_performance_comparison.png bytes= 102447
```






```
saved figure: figures_factor_graphical/17_feature_importance_graph_augmented_classifier.png bytes= 122285
```

	warning_rank	date	next_date	hs_code	hs_name	country_code	country_name	import_value	import_weight	unit_pri
0	1	2024-01-01	2024-02-01	0202	쇠고기(냉동한 것으로 한정한다)	US	미국	190561916	24349110	7.8262
1	2	2024-08-01	2024-09-01	8501	전동기와 발전기(발전세트는 제외한다)	PH	필리핀	110540249	67023	1649.2882
2	3	2024-10-01	2024-11-01	8501	전동기와 발전기(발전세트는 제외한다)	PH	필리핀	103275718	61829	1670.3443
3	4	2024-04-01	2024-05-01	2709	석유와 역청유(瀝青油)(원유로 한정한다)	MX	멕시코	150453057	288365237	0.5217
4	5	2024-07-01	2024-08-01	2709	석유와 역청유(瀝青油)(원유로 한정한다)	MX	멕시코	282645526	493392648	0.5728
5	6	2024-09-01	2024-10-01	8501	전동기와 발전기(발전세트는 제외한다)	PH	필리핀	103436367	68029	1520.4746
6	7	2024-10-01	2024-11-01	2709	석유와 역청유(瀝青油)(원유로 한정한다)	IQ	이라크	564765082	1107082746	0.5101
7	8	2024-07-01	2024-08-01	2607	납광과 그 정광(精鑛)	MX	멕시코	173078223	18624910	9.2928
8	9	2024-05-01	2024-06-01	8541	반도체 디바이스(예: 다이오드 · 트랜지스터 · 반도체 기반 트랜스듀서), 감광성 반도체 ...	CN	중국	143994539	14156625	10.1715
9	10	2024-04-01	2024-05-01	8541	반도체 디바이스(예: 다이오드 · 트랜지스터 · 반도체 기반 트랜스듀서), 감광성 반도체 ...	CN	중국	134808001	12902041	10.4485
10	11	2024-10-01	2024-11-01	8541	반도체 디바이스(예: 다이오드 · 트랜지스터 · 반도체 기반 트랜스듀서), 감광성 반도체 ...	CN	중국	134346616	22280152	6.0298
11	12	2024-11-01	2024-12-01	2607	납광과 그 정광(精鑛)	MX	멕시코	81822826	9499647	8.6132
12	13	2024-09-01	2024-10-01	2709	석유와 역청유(瀝青油)(원유로 한정한다)	CA	캐나다	61180147	122832093	0.4980
13	14	2024-11-01	2024-12-01	2709	석유와 역청유(瀝青油)(원유로 한정한다)	IQ	이라크	578863322	1122429548	0.5157
14	15	2024-02-01	2024-03-01	2709	석유와 역청유(瀝青油)(원유로 한정한다)	IQ	이라크	659058690	1216873555	0.5416
15	16	2024-03-01	2024-04-01	8541	반도체 디바이스(예: 다이오드 · 트랜지스터 · 반도체 기반 트랜스듀서), 감광성 반도체 ...	CN	중국	129286540	12026228	10.7503

16	17	2024-08-01	2024-09-01	8541	반도체 디바이스(예: 다이오드 · 트랜지스터 · 반도체 기반 트랜스듀서), 감광성 반도체 ...	CN	중국	119819661	10957410	10.9350
17	18	2024-11-01	2024-12-01	8541	반도체 디바이스(예: 다이오드 · 트랜지스터 · 반도체 기반 트랜스듀서), 감광성 반도체 ...	CN	중국	132437474	23663711	5.5966
18	19	2024-02-01	2024-03-01	8541	반도체 디바이스(예: 다이오드 · 트랜지스터 · 반도체 기반 트랜스듀서), 감광성 반도체 ...	CN	중국	113197221	9474187	11.9479
19	20	2024-09-01	2024-10-01	8541	반도체 디바이스(예: 다이오드 · 트랜지스터 · 반도체 기반 트랜스듀서), 감광성 반도체 ...	CN	중국	116393034	12927838	9.0032
20	21	2024-07-01	2024-08-01	8541	반도체 디바이스(예: 다이오드 · 트랜지스터 · 반도체 기반 트랜스듀서), 감광성 반도체 ...	CN	중국	132556721	12505950	10.5994
21	22	2024-01-01	2024-02-01	8541	반도체 디바이스(예: 다이오드 · 트랜지스터 · 반도체 기반 트랜스듀서), 감광성 반도체 ...	CN	중국	169729155	13474225	12.5965
22	23	2024-06-01	2024-07-01	8541	반도체 디바이스(예: 다이오드 · 트랜지스터 · 반도체 기반 트랜스듀서), 감광성 반도체 ...	CN	중국	129221040	12549234	10.2971
23	24	2024-05-01	2024-06-01	2607	납광과 그 정광(精鑛)	MX	멕시코	64752017	9293851	6.9671
24	25	2024-03-01	2024-04-01	8708	부분품과 부속품(제8701호부터 제8705호까지의 차량용으로 한정한다)	CN	중국	177641447	24718886	7.1864
25	26	2024-06-01	2024-07-01	8703	주로 사람을 수송할 수 있도록 설계된 승용자동차와 그 밖의 차량[제8702호의 것은...	GB	영국	83303327	2042652	40.7819
26	27	2024-04-01	2024-05-01	8708	부분품과 부속품(제8701호부터 제8705호까지의 차량용으로 한정한다)	CN	중국	196904612	26629093	7.3943
		2024-			전동기와 발전기(발전세					

27	28	2024-01-01	2024-02-01	8501	단위를 전제트는 제외한다)	PH	필리핀	58861234	40763	1443.9868
28	29	2024-04-01	2024-05-01	8473	제8470호부터 제8472호까지에 해당하는 기계에 전용되거나 주로 사용되는 부분품과...	CN	중국	199907164	1733569	115.3153
29	30	2024-07-01	2024-08-01	8703	주로 사람을 수송할 수 있도록 설계된 승용자동차와 그 밖의 차량[제8702호의 것은...	GB	영국	83997356	2087311	40.2418
30	31	2024-01-01	2024-02-01	8708	부분품과 부속품(제8701호부터 제8705호까지의 차량용으로 한정한다)	CN	중국	212669114	29919432	7.1080
31	32	2024-07-01	2024-08-01	7308	철강으로 만든 구조물(제9406호의 조립식 건축물은 제외한다)과 구조물의 부분품[예...	CN	중국	137028852	89322025	1.5340
32	33	2024-07-01	2024-08-01	7112	귀금속이나 귀금속을 입힌 금속의 웨이스트(waste)와 스크랩(scrap), 귀금속...	AU	호주	59831676	86350	692.8972
33	34	2024-05-01	2024-06-01	8708	부분품과 부속품(제8701호부터 제8705호까지의 차량용으로 한정한다)	CN	중국	202306373	27719138	7.2984
34	35	2024-06-01	2024-07-01	8708	부분품과 부속품(제8701호부터 제8705호까지의 차량용으로 한정한다)	CN	중국	187803191	25692586	7.3096
35	36	2024-07-01	2024-08-01	8501	전동기와 발전기(발전세트는 제외한다)	PH	필리핀	56331878	57243	984.0832
36	37	2024-06-01	2024-07-01	7308	철강으로 만든 구조물(제9406호의 조립식 건축물은 제외한다)과 구조물의 부분품[예...	CN	중국	135109100	86728118	1.5578
37	38	2024-11-01	2024-12-01	8708	부분품과 부속품(제8701호부터 제8705호까지의 차량용으로 한정한다)	CN	중국	219218743	27408389	7.9982
38	39	2024-04-01	2024-05-01	2607	납광과 그 정광(精鑛)	MX	멕시코	51074633	8356650	6.1118
39	40	2024-07-01	2024-08-01	8708	부분품과 부속품(제8701호부터 제8705호까지의 차량용	CN	중국	198107650	25986111	7.6235

```
# %% [markdown]
# ## Regime-switching residual graphical networks
#
# 기존 static network는 2020~2024 전체 기간의 평균적인 조건부 의존 구조를 보여준다.
# 여기서는 기간을 나눠서 공통요인 제거 후 residual network가 시기별로 어떻게 달라지는지 본다.
#
# 핵심 질문:
# - 코로나 시기, 공급망 위기 시기, 정상화 시기에 HS 간 residual dependency가 달라지는가?
# - 특정 시기에 네트워크가 더 dense해지는가?
# - 중심 HS가 시기별로 바뀌는가?
#
# 사용 데이터:
# residual_mat: PCA 공통요인 제거 후 HS별 잔차 u_{h,t}
#
# Regime별 모델:
# u_t^{(r)} ~ N(0, Sigma_u^{(r)})
# Omega_u^{(r)} = Sigma_u^{(r), -1}
#
# Omega_{ij}^{(r)} != 0이면 regime r에서 HS i,j가 조건부 의존.

# %%
from sklearn.covariance import GraphicalLasso
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import networkx as nx
from pathlib import Path

Path("figures_regime_network").mkdir(exist_ok=True)
Path("output_regime_network").mkdir(exist_ok=True)

# ----------------------------
# Regime settings
# ----------------------------
REGIMES = {
    "COVID_2020": ("2020-01-01", "2020-12-31"),
    "SupplyChain_2021_2022": ("2021-01-01", "2022-12-31"),
    "Normalization_2023_2024": ("2023-01-01", "2024-12-31"),
}

# 너무 많은 node를 regime별로 쓰면 T가 작아서 불안정하다.
# 120개 전체보다 수입금액 상위 50개 정도가 안정적이다.
REGIME_MAX_HS_NODES = 50

# Global Graphical LASSO alpha가 있으면 그것을 기준으로 사용.
# Regime별 표본 수가 작으므로 약간 더 보수적으로 키운다.
REGIME_ALPHA_MULTIPLIER = 1.25

# 시각화용 최소 partial correlation threshold.
# 너무 sparse하면 0.02~0.03, 너무 dense하면 0.05 이상으로 조정.
REGIME_EDGE_THRESHOLD = 0.03

# 각 regime 그림에 최대 표시 edge 수.
REGIME_TOP_EDGES_TO_PLOT = 80

# bootstrap은 시간이 오래 걸리므로 일단 False.
RUN_REGIME_BOOTSTRAP = False
REGIME_BOOTSTRAP_B = 50
REGIME_BOOTSTRAP_BLOCK_SIZE = 4

print("Regime network settings ready.")
```

다)

```
Regime network settings ready.
```

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```
# %% [markdown]
# ## Helper functions for regime-specific Graphical LASSO
#
# 이 셀은 regime별 residual matrix를 받아서:
# 1. 표준화
# 2. Graphical LASSO 적합
# 3. partial correlation matrix 계산
# 4. edge table 생성
# 5. centrality 계산
# 을 수행한다.

# %%
def choose_regime_nodes(hs_month_sel, selected_hs, max_nodes=50):
    """
    전체 selected_hs 중에서 수입금액 기준 상위 max_nodes개만 regime network node로 선택한다.
    """
```

```
    import_rank = (
        hs_month_sel[hs_month_sel["hs_code"].astype(str).isin([str(x) for x in selected_hs])]
        .groupby("hs_code")["hs_import_value"]
        .sum()
        .sort_values(ascending=False)
    )
    nodes = import_rank.head(max_nodes).index.astype(str).tolist()
    return nodes


def standardize_regime_matrix(X_df):
    """
    regime 내부에서 column-wise standardization.
    결측 또는 0 variance는 0으로 채운다.
    """
    X = X_df.copy().astype(float)
    mu = X.mean(axis=0, skipna=True)
    sd = X.std(axis=0, skipna=True).replace(0, np.nan)
    Z = (X - mu) / sd
    Z = Z.replace([np.inf, -np.inf], np.nan).fillna(0.0)
    return Z


def precision_to_partial_corr(precision):
    """
    Precision matrix Omega를 partial correlation matrix로 변환.
    rho_ij = - Omega_ij / sqrt(Omega_ii Omega_jj)
    """
    diag = np.sqrt(np.diag(precision))
    pc = -precision / np.outer(diag, diag)
    np.fill_diagonal(pc, 1.0)
    return pc


def build_edge_table_from_partial(pc_df, threshold=0.03):
    """
    partial correlation matrix에서 edge table 생성.
    """
    hs_codes = list(pc_df.index)
    rows = []
    for i in range(len(hs_codes)):
        for j in range(i + 1, len(hs_codes)):
            pc = float(pc_df.iloc[i, j])
            if np.isfinite(pc) and abs(pc) >= threshold:
                rows.append({
                    "hs_i": hs_codes[i],
                    "hs_j": hs_codes[j],
                    "partial_corr": pc,
                    "abs_partial_corr": abs(pc),
                    "sign": "positive" if pc > 0 else "negative",
                })
    edge_df = pd.DataFrame(rows)
    if len(edge_df) > 0:
        edge_df = edge_df.sort_values("abs_partial_corr", ascending=False).reset_index(drop=True)
    return edge_df


def edge_table_to_graph(edge_df):
    """
    edge table을 networkx Graph로 변환.
    """
    G = nx.Graph()
    if edge_df is None or len(edge_df) == 0:
        return G

    for _, r in edge_df.iterrows():
        G.add_edge(
            str(r["hs_i"]),
            str(r["hs_j"]),
            weight=float(r["abs_partial_corr"]),
            partial_corr=float(r["partial_corr"]),
            sign=str(r["sign"]),
        )
    return G


def compute_graph_centrality(G, regime_name, node_info=None):
    """
    regime graph의 centrality table 생성.
    """
    if G.number_of_nodes() == 0:
        return pd.DataFrame(columns=[
```

```
            "regime", "hs_code", "degree", "weighted_degree",
            "betweenness", "component_id", "component_size",
            "mean_Z", "total_import"
        ])

    weighted_degree = dict(G.degree(weight="weight"))
    degree = dict(G.degree())
    betweenness = nx.betweenness_centrality(G, weight="weight") if G.number_of_edges() > 0 else {n: 0 for n in G.nodes()}

    components = list(nx.connected_components(G))
    comp_map = {}
    comp_size = {}
    for k, comp in enumerate(components):
        for node in comp:
            comp_map[node] = k
            comp_size[node] = len(comp)

    rows = []
    for n in G.nodes():
        rows.append({
            "regime": regime_name,
            "hs_code": n,
            "degree": degree.get(n, 0),
            "weighted_degree": weighted_degree.get(n, 0.0),
            "betweenness": betweenness.get(n, 0.0),
            "component_id": comp_map.get(n, -1),
            "component_size": comp_size.get(n, 1),
            "mean_Z": node_info.get(n, {}).get("mean_Z", np.nan) if node_info else np.nan,
            "total_import": node_info.get(n, {}).get("total_import", np.nan) if node_info else np.nan,
        })

    return pd.DataFrame(rows).sort_values(["weighted_degree", "degree"], ascending=False)


def fit_regime_graphical_lasso(
    residual_mat,
    regime_name,
    start_date,
    end_date,
    nodes,
    alpha,
    threshold=0.03,
):
    """
    특정 regime 기간에 대해 residual Graphical LASSO를 적합한다.
    """
    start = pd.to_datetime(start_date)
    end = pd.to_datetime(end_date)

    U_regime = residual_mat.copy()
    U_regime.index = pd.to_datetime(U_regime.index)
    U_regime = U_regime.loc[(U_regime.index >= start) & (U_regime.index <= end), nodes]

    if U_regime.shape[0] < 8:
        print(f"[WARNING] {regime_name}: too few months, n={U_regime.shape[0]}")

    U_std = standardize_regime_matrix(U_regime)

    gl = GraphicalLasso(alpha=alpha, max_iter=2000)
    gl.fit(U_std.values)

    precision = gl.precision_
    partial_corr = precision_to_partial_corr(precision)

    pc_df = pd.DataFrame(partial_corr, index=nodes, columns=nodes)
    edge_df = build_edge_table_from_partial(pc_df, threshold=threshold)
    G = edge_table_to_graph(edge_df)

    result = {
        "regime": regime_name,
        "start_date": start,
        "end_date": end,
        "n_months": U_std.shape[0],
        "n_nodes": len(nodes),
        "alpha": alpha,
        "U_std": U_std,
        "precision": precision,
        "partial_corr_df": pc_df,
        "edge_df": edge_df,
        "graph": G,
    }
    return result
```

```
# Select regime nodes
regime_nodes = choose_regime_nodes(
    hs_month_sel=hs_month_sel,
    selected_hs=selected_hs,
    max_nodes=REGIME_MAX_HS_NODES
)

print("Regime nodes:", len(regime_nodes))
print(regime_nodes[:20])

# Node info for plotting
node_info_regime = (
    hs_month_sel[hs_month_sel["hs_code"].astype(str).isin(regime_nodes)]
    .groupby("hs_code")
    .agg(
        mean_Z=("Z_ht", "mean"),
        total_import=("hs_import_value", "sum"),
        n_month=("date", "nunique"),
    )
    .to_dict("index")
)

# Pick alpha
if "alpha_selected" in globals():
    regime_alpha = float(alpha_selected) * REGIME_ALPHA_MULTIPLIER
else:
    regime_alpha = 0.10 * REGIME_ALPHA_MULTIPLIER

print("Regime Graphical LASSO alpha:", regime_alpha)
```

```
Regime nodes: 50
['2709', '8542', '2711', '2710', '8486', '2701', '8703', '8517', '2601', '8471', '2825', '8541', '2603', '8473', '8507', '3004', '8708',
Regime Graphical LASSO alpha: 0.5228568984900968
```

```
# %% [markdown]
# ## Fit regime-specific residual graphical networks
#
# 각 regime에 대해 Graphical LASSO를 따로 적합한다.
# 결과는 output_regime_network/ 아래에 저장된다.

# %%
regime_results = {}

for regime_name, (start_date, end_date) in REGIMES.items():
    print("Wn" + "=" * 80)
    print(f"Fitting regime: {regime_name} | {start_date} to {end_date}")

    res = fit_regime_graphical_lasso(
        residual_mat=residual_mat,
        regime_name=regime_name,
        start_date=start_date,
        end_date=end_date,
        nodes=regime_nodes,
        alpha=regime_alpha,
        threshold=REGIME_EDGE_THRESHOLD,
    )

    regime_results[regime_name] = res

    # Save matrices and edges
    res["partial_corr_df"].to_csv(
        f"output_regime_network/{regime_name}_partial_corr.csv",
        encoding="utf-8-sig"
    )

    if len(res["edge_df"]) > 0:
        res["edge_df"].to_csv(
            f"output_regime_network/{regime_name}_edges.csv",
            index=False,
            encoding="utf-8-sig"
        )

    print("n_months:", res["n_months"])
    print("n_nodes:", res["n_nodes"])
    print("n_edges:", len(res["edge_df"]))
    display(res["edge_df"].head(15) if len(res["edge_df"]) > 0 else pd.DataFrame())
```

```
============================================================================
Fitting regime: COVID_2020 | 2020-01-01 to 2020-12-31
n_months: 12
n_nodes: 50
n_edges: 63
```

	hs_i	hs_j	partial_corr	abs_partial_corr	sign
0	2710	8901	-0.268636	0.268636	negative
1	8534	7403	0.255504	0.255504	positive
2	8541	7204	-0.249532	0.249532	negative
3	8517	8471	0.244860	0.244860	positive
4	4202	8534	-0.219160	0.219160	negative
5	2709	1005	-0.198517	0.198517	negative
6	3002	8544	-0.196541	0.196541	negative
7	3002	7601	0.196264	0.196264	positive
8	2825	8802	0.189199	0.189199	positive
9	8473	0202	0.182243	0.182243	positive
10	9018	8901	0.177722	0.177722	positive
11	7208	8528	-0.174537	0.174537	negative
12	7208	8802	0.171303	0.171303	positive
13	8528	2607	-0.170272	0.170272	negative
14	1005	7326	-0.160206	0.160206	negative

```
============================================================================
Fitting regime: SupplyChain_2021_2022 | 2021-01-01 to 2022-12-31
n_months: 24
n_nodes: 50
n_edges: 2
```

	hs_i	hs_j	partial_corr	abs_partial_corr	sign
0	2710	8544	0.089556	0.089556	positive
1	3004	8411	0.087409	0.087409	positive

```
============================================================================
Fitting regime: Normalization_2023_2024 | 2023-01-01 to 2024-12-31
n_months: 24
n_nodes: 50
n_edges: 9
```

	hs_i	hs_j	partial_corr	abs_partial_corr	sign
0	8523	3818	0.151171	0.151171	positive
1	8411	7204	-0.135432	0.135432	negative
2	7601	9018	0.128697	0.128697	positive
3	2603	8529	0.091032	0.091032	positive
4	8486	8501	-0.077791	0.077791	negative
5	8529	0202	0.059532	0.059532	positive
6	8486	8529	0.051748	0.051748	positive
7	2710	8536	0.043415	0.043415	positive
8	2711	3002	-0.031538	0.031538	negative

```
# %% [markdown]
# ## Regime network summary metrics
#
# 시기별 네트워크가 얼마나 조밀한지, 중심성이 어떻게 달라지는지 요약한다.

# %%
summary_rows = []
centrality_tables = []

possible_edges = len(regime_nodes) * (len(regime_nodes) - 1) / 2

for regime_name, res in regime_results.items():
    G = res["graph"]
    edge_df = res["edge_df"]
```

```
        n_edges = G.number_of_edges()
        n_nodes_connected = G.number_of_nodes()
        density = n_edges / possible_edges if possible_edges > 0 else np.nan

        if len(edge_df) > 0:
            mean_abs_pc = edge_df["abs_partial_corr"].mean()
            max_abs_pc = edge_df["abs_partial_corr"].max()
            positive_share = (edge_df["partial_corr"] > 0).mean()
        else:
            mean_abs_pc = 0.0
            max_abs_pc = 0.0
            positive_share = np.nan

        components = list(nx.connected_components(G)) if G.number_of_nodes() > 0 else []
        largest_component_size = max([len(c) for c in components]) if components else 0

        cent = compute_graph_centrality(G, regime_name, node_info=node_info_regime)
        centrality_tables.append(cent)

        top_node = cent.iloc[0]["hs_code"] if len(cent) > 0 else None
        top_wdegree = cent.iloc[0]["weighted_degree"] if len(cent) > 0 else 0.0

        summary_rows.append({
            "regime": regime_name,
            "start_date": res["start_date"],
            "end_date": res["end_date"],
            "n_months": res["n_months"],
            "n_nodes_total": len(regime_nodes),
            "n_nodes_connected": n_nodes_connected,
            "n_edges": n_edges,
            "edge_density": density,
            "mean_abs_partial_corr": mean_abs_pc,
            "max_abs_partial_corr": max_abs_pc,
            "positive_edge_share": positive_share,
            "n_components": len(components),
            "largest_component_size": largest_component_size,
            "top_central_hs": top_node,
            "top_weighted_degree": top_wdegree,
        })

    regime_summary = pd.DataFrame(summary_rows)
    regime_centrality = pd.concat(centrality_tables, ignore_index=True) if centrality_tables else pd.DataFrame()

    regime_summary.to_csv("output_regime_network/regime_network_summary.csv", index=False, encoding="utf-8-sig")
    regime_centrality.to_csv("output_regime_network/regime_network_centrality.csv", index=False, encoding="utf-8-sig")

    display(regime_summary)
    display(regime_centrality.head(30))
```

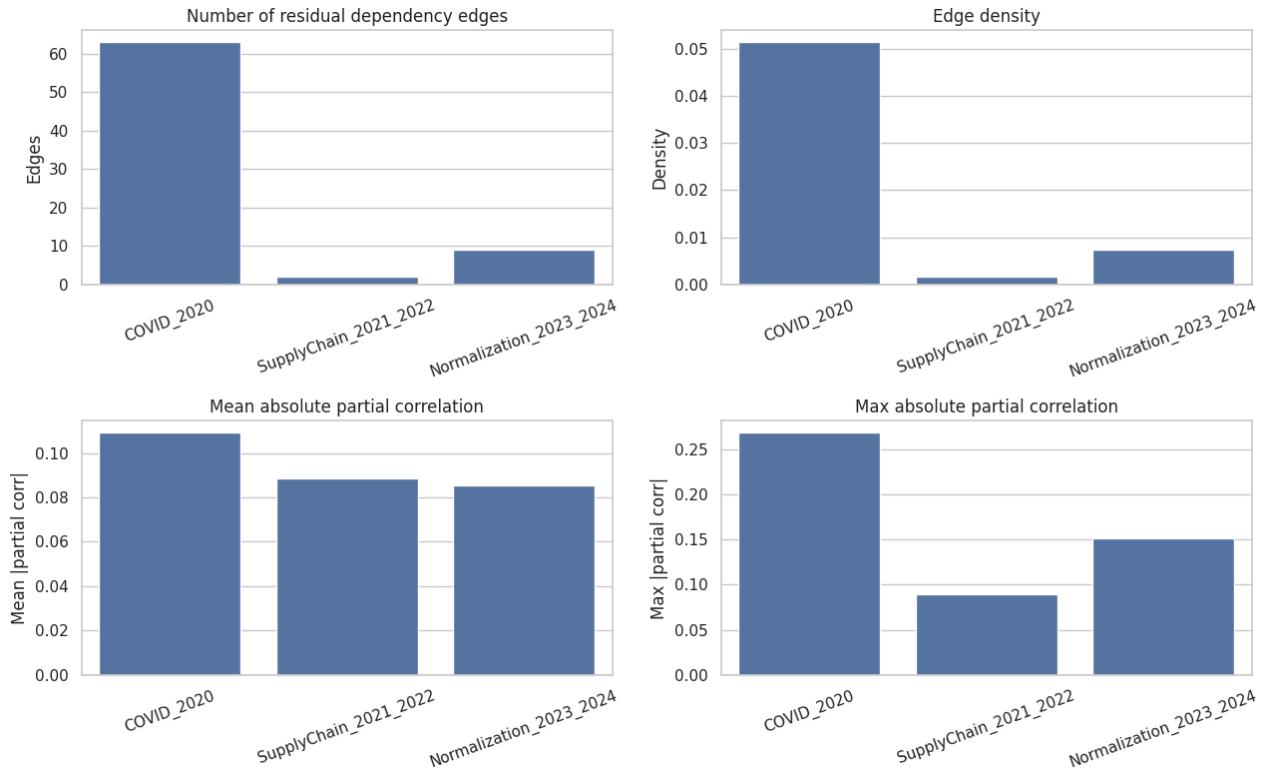
	regime	start_date	end_date	n_months	n_nodes_total	n_nodes_connected	n_edges	edge_density	mean_
0	COVID_2020	2020-01-01	2020-12-31	12	50	45	63	0.051429	
1	SupplyChain_2021_2022	2021-01-01	2022-12-31	24	50	4	2	0.001633	
2	Normalization_2023_2024	2023-01-01	2024-12-31	24	50	15	9	0.007347	

	regime	hs_code	degree	weighted_degree	betweenness	component_id	component_size	mean_Z	total_import
0	COVID_2020	3002	6	0.755027	0.096195	0	40	1.131263	20400993399
1	COVID_2020	2709	7	0.644441	0.188161	0	40	0.952461	388924477700
2	COVID_2020	4202	5	0.621575	0.135307	0	40	2.061739	17612069604
3	COVID_2020	1005	5	0.620513	0.169133	0	40	0.868983	16307891254
4	COVID_2020	8901	4	0.587186	0.040169	0	40	1.132927	11313957768
5	COVID_2020	8528	5	0.555962	0.199789	0	40	1.475545	10756013118
6	COVID_2020	7601	5	0.517766	0.031712	0	40	0.475198	18879017193
7	COVID_2020	7204	4	0.502506	0.084567	0	40	0.739702	9968521732
8	COVID_2020	8534	2	0.474664	0.000000	0	40	1.207470	14811256378
9	COVID_2020	8802	3	0.468127	0.000000	0	40	3.489594	9438287688
10	COVID_2020	7208	3	0.460312	0.058140	0	40	1.644493	15995286621
11	COVID_2020	2825	5	0.455132	0.140592	0	40	0.368367	28681519637
12	COVID_2020	7326	5	0.435389	0.135307	0	40	0.711825	10125222406
13	COVID_2020	2710	3	0.421213	0.093023	0	40	0.581579	110320867308
14	COVID_2020	2607	4	0.405027	0.153277	0	40	3.082629	9614405173
15	COVID_2020	8501	4	0.376600	0.100423	0	40	1.616694	17149652043
16	COVID_2020	8544	4	0.351556	0.143763	0	40	1.454508	19429824885
17	COVID_2020	8517	3	0.337624	0.104651	0	40	0.740962	58703128127
18	COVID_2020	8471	2	0.330713	0.022199	0	40	0.886357	42672741406
19	COVID_2020	8541	2	0.330659	0.000000	0	40	2.042224	27864426089
20	COVID_2020	8703	4	0.326632	0.142706	0	40	1.197428	65652543941
21	COVID_2020	7403	3	0.324975	0.086681	0	40	0.508137	12395423807
22	COVID_2020	7404	3	0.318490	0.036998	0	40	1.405435	11896904870
23	COVID_2020	9018	3	0.284927	0.114165	0	40	0.720514	12620783980
24	COVID_2020	8486	3	0.239004	0.049683	0	40	1.035292	93835722545
25	COVID_2020	9031	2	0.217026	0.001057	2	3	1.004377	19836560433
26	COVID_2020	2711	2	0.200733	0.032770	0	40	0.689981	182381188096
27	COVID_2020	8523	3	0.198323	0.050740	0	40	1.088571	13407404679
28	COVID_2020	8473	1	0.182243	0.000000	1	2	1.890306	27463176207
29	COVID_2020	0202	1	0.182243	0.000000	1	2	1.831347	11714789195

다음 단계: regime_summary 변수로 코드 생성 New interactive sheet

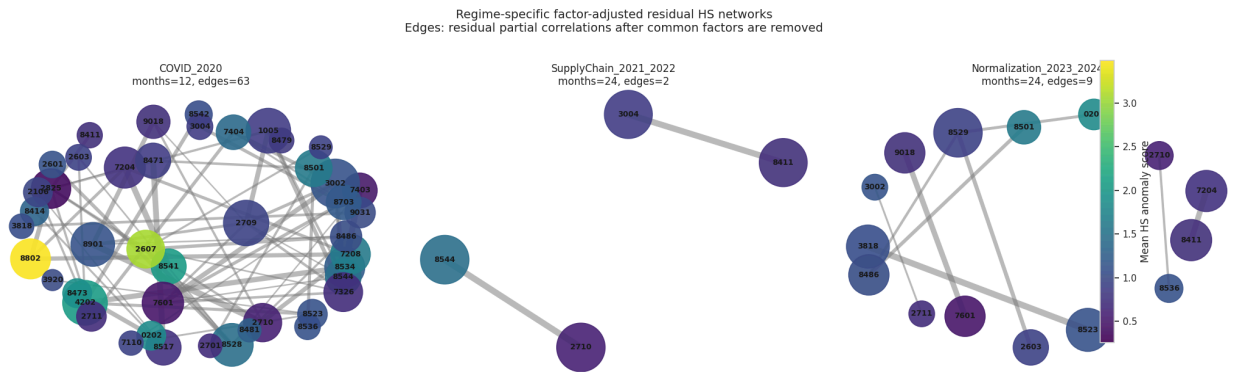
```
# %% [markdown]
# ## Visualize regime-level network metrics
#
# 시기별 edge 수, density, 평균 partial correlation, 최대 partial correlation을 비교한다.

# %%
fig, axes = plt.subplots(2, 2, figsize=(13, 8))

sns.barplot(data=regime_summary, x="regime", y="n_edges", ax=axes[0, 0])
axes[0, 0].set_title("Number of residual dependency edges")
axes[0, 0].set_xlabel("")
axes[0, 0].set_ylabel("Edges")
axes[0, 0].tick_params(axis="x", rotation=20)

sns.barplot(data=regime_summary, x="regime", y="edge_density", ax=axes[0, 1])
axes[0, 1].set_title("Edge density")
```

```
axes[0, 1].set_xlabel("")
axes[0, 1].set_ylabel("Density")
axes[0, 1].tick_params(axis="x", rotation=20)

sns.barplot(data=regime_summary, x="regime", y="mean_abs_partial_corr", ax=axes[1, 0])
axes[1, 0].set_title("Mean absolute partial correlation")
axes[1, 0].set_xlabel("")
axes[1, 0].set_ylabel("Mean |partial corr|")
axes[1, 0].tick_params(axis="x", rotation=20)

sns.barplot(data=regime_summary, x="regime", y="max_abs_partial_corr", ax=axes[1, 1])
axes[1, 1].set_title("Max absolute partial correlation")
axes[1, 1].set_xlabel("")
axes[1, 1].set_ylabel("Max |partial corr|")
axes[1, 1].tick_params(axis="x", rotation=20)

plt.tight_layout()
plt.savefig("figures_regime_network/01_regime_network_metrics.png", dpi=160, bbox_inches="tight")
plt.show()
```




```
# %% [markdown]
# ## Plot regime-specific residual conditional-dependence networks
#
# 세 regime의 residual conditional-dependence network를 나란히 비교한다.
#
# Node size: weighted degree
# Node color: mean HS anomaly score
# Edge width: absolute residual partial correlation

# %%
def plot_regime_network_panel(regime_results, node_info, max_edges=80):
    n_regimes = len(regime_results)
    fig, axes = plt.subplots(1, n_regimes, figsize=(7 * n_regimes, 6))

    if n_regimes == 1:
        axes = [axes]

    # 공통 color scale
    mean_z_values = [node_info.get(n, {}).get("mean_Z", np.nan) for n in regime_nodes]
    vmin = np.nanmin(mean_z_values)
```

```
    vmax = np.nanmax(mean_z_values)

    for ax, (regime_name, res) in zip(axes, regime_results.items()):
        edge_df = res["edge_df"].copy()

        if len(edge_df) > max_edges:
            edge_df = edge_df.sort_values("abs_partial_corr", ascending=False).head(max_edges)

        G = edge_table_to_graph(edge_df)

        if G.number_of_edges() == 0:
            ax.text(0.5, 0.5, "No edges", ha="center", va="center", fontsize=14)
            ax.set_title(f"{regime_name}₩nNo residual edges")
            ax.axis("off")
            continue

        pos = nx.spring_layout(G, seed=123, k=0.9)

        wdeg = dict(G.degree(weight="weight"))
        max_wdeg = max(wdeg.values()) if wdeg else 1.0
        node_sizes = [500 + 2800 * wdeg.get(n, 0.0) / max_wdeg for n in G.nodes()]

        node_colors = [node_info.get(n, {}).get("mean_Z", np.nan) for n in G.nodes()]

        edge_weights = [G[u][v]["weight"] for u, v in G.edges()]
        max_edge_w = max(edge_weights) if edge_weights else 1.0
        edge_widths = [0.8 + 6.5 * w / max_edge_w for w in edge_weights]

        nx.draw_networkx_edges(
            G,
            pos,
            ax=ax,
            width=edge_widths,
            edge_color="gray",
            alpha=0.55,
        )

        nodes = nx.draw_networkx_nodes(
            G,
            pos,
            ax=ax,
            node_size=node_sizes,
            node_color=node_colors,
            cmap="viridis",
            vmin=vmin,
            vmax=vmax,
            alpha=0.9,
        )

        nx.draw_networkx_labels(
            G,
            pos,
            ax=ax,
            font_size=9,
            font_weight="bold",
        )

        ax.set_title(
            f"{regime_name}₩n"
            f"months={res['n_months']}, edges={G.number_of_edges()}"
        )
        ax.axis("off")

    cbar = fig.colorbar(nodes, ax=axes, fraction=0.025, pad=0.02)
    cbar.set_label("Mean HS anomaly score")

    plt.suptitle(
        "Regime-specific factor-adjusted residual HS networks₩n"
        "Edges: residual partial correlations after common factors are removed",
        y=1.02,
        fontsize=14,
    )
    plt.tight_layout()
    plt.savefig("figures_regime_network/02_regime_networks_panel.png", dpi=160, bbox_inches="tight")
    plt.show()


plot_regime_network_panel(regime_results, node_info_regime, max_edges=REGIME_TOP_EDGES_TO_PLOT)
```




```
# %% [markdown]
# ## Top residual dependency edges by regime
#
# 시기별로 어떤 HS pair가 강하게 연결되는지 비교한다.

# %%
top_edge_tables = []

for regime_name, res in regime_results.items():
    edge_df = res["edge_df"].copy()
    if len(edge_df) == 0:
        continue

    edge_df["regime"] = regime_name
    edge_df["edge"] = edge_df.apply(lambda r: "-".join(sorted([str(r["hs_i"]), str(r["hs_j"])])), axis=1)
    edge_df["rank_in_regime"] = np.arange(1, len(edge_df) + 1)
    top_edge_tables.append(edge_df.head(20))

regime_top_edges = pd.concat(top_edge_tables, ignore_index=True) if top_edge_tables else pd.DataFrame()
regime_top_edges.to_csv("output_regime_network/regime_top_edges.csv", index=False, encoding="utf-8-sig")

display(regime_top_edges)

# Plot top 10 edges per regime
if len(regime_top_edges) > 0:
    plot_edges = regime_top_edges[regime_top_edges["rank_in_regime"] <= 10].copy()
    plot_edges["edge_label"] = plot_edges["edge"] + " (" + plot_edges["regime"] + ")"

    plt.figure(figsize=(10, max(5, 0.35 * len(plot_edges))))
    sns.barplot(
        data=plot_edges.sort_values("abs_partial_corr", ascending=True),
        y="edge_label",
        x="abs_partial_corr",
        hue="regime",
        dodge=False,
    )
    plt.title("Top residual dependency edges by regime")
    plt.xlabel("|Residual partial correlation|")
    plt.ylabel("HS pair")
    plt.legend(title="Regime", loc="lower right")
    plt.tight_layout()
    plt.savefig("figures_regime_network/03_top_edges_by_regime.png", dpi=160, bbox_inches="tight")
    plt.show()
```

	hs_i	hs_j	partial_corr	abs_partial_corr	sign	regime	edge	rank_in_regime
0	2710	8901	-0.268636	0.268636	negative	COVID_2020	2710-8901	1
1	8534	7403	0.255504	0.255504	positive	COVID_2020	7403-8534	2
2	8541	7204	-0.249532	0.249532	negative	COVID_2020	7204-8541	3
3	8517	8471	0.244860	0.244860	positive	COVID_2020	8471-8517	4
4	4202	8534	-0.219160	0.219160	negative	COVID_2020	4202-8534	5
5	2709	1005	-0.198517	0.198517	negative	COVID_2020	1005-2709	6
6	3002	8544	-0.196541	0.196541	negative	COVID_2020	3002-8544	7